

Factorising Quadratics

Complete Worked Solutions with Diagrams

Wisest Maths — Year 1 A-Level (topic ref: qc1)

Generated automatically from the question bank

About this document

This booklet contains every question from the **Factorising Quadratics** bank (qc1-001 to qc1-070), each presented with a full step-by-step worked solution, a check by expansion, and supporting diagrams drawn in TikZ:

- an **area / box model**, where every cell is the product of its row and column term — the four (or more) cells sum to the original quadratic; and
- for numeric examples, a **graph** $y = f(x)$ showing the parabola together with its roots (where it crosses the x -axis) and its vertex.

The box model makes the link between *factorising* (splitting a rectangle into pieces) and *expanding* (recombining them) explicit, while the graph connects the algebra to the geometry of the curve.

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Question 1 qc1-001

Factorise: $x^2 + 5x + 6$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Write down two brackets, each starting with x .

$$(x \quad)(x \quad)$$

Since the leading coefficient is 1, both brackets begin with x .

Step 2. Find two numbers that multiply to give +6 and add to give +5.

$$2 \times 3 = 6, \quad 2 + 3 = 5$$

The pair (2,3) works for both conditions.

Step 3. Place the numbers into the brackets.

$$(x + 2)(x + 3)$$

Both signs are positive because c and b are positive.

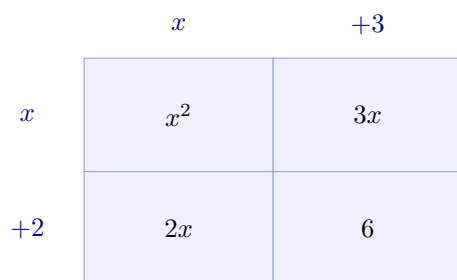
Final answer: $(x + 2)(x + 3)$

Check by expanding

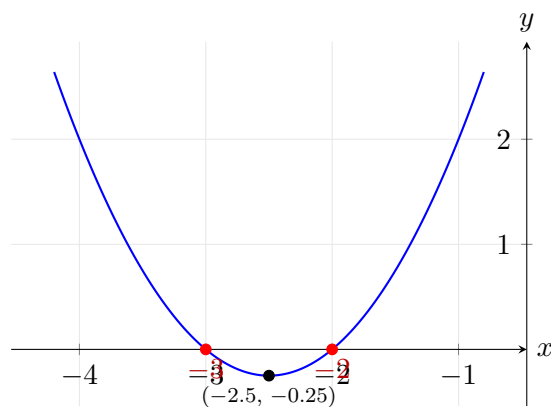
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 3x + 2x + 6$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 2 qc1-002

Factorise: $x^2 + 7x + 12$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up two brackets starting with x .

$$(x \quad)(x \quad)$$

The leading term is x^2 , so both brackets begin with x .

Step 2. Find two numbers that multiply to 12 and add to 7.

$$3 \times 4 = 12, \quad 3 + 4 = 7$$

Factor pairs of 12 are $(1,12)$, $(2,6)$, $(3,4)$. Only $(3,4)$ adds to 7.

Step 3. Write the factorised form.

$$(x + 3)(x + 4)$$

Both signs are positive.

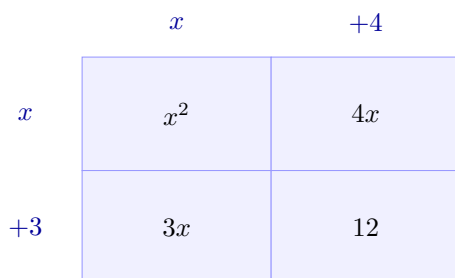
Final answer: $(x + 3)(x + 4)$

Check by expanding

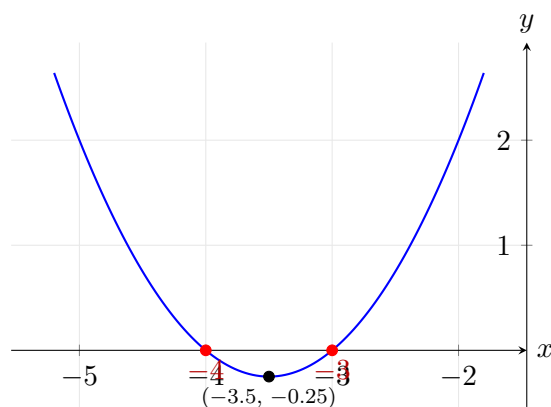
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 4x + 3x + 12$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 3 qc1-003

Factorise: $x^2 + 9x + 20$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up the two brackets.

$$(x \quad)(x \quad)$$

Leading coefficient is 1.

Step 2. Find two numbers multiplying to 20 and adding to 9.

$$4 \times 5 = 20, \quad 4 + 5 = 9$$

Factor pairs of 20: (1,20), (2,10), (4,5). Only (4,5) adds to 9.

Step 3. Write the factors.

$$(x + 4)(x + 5)$$

Both positive.

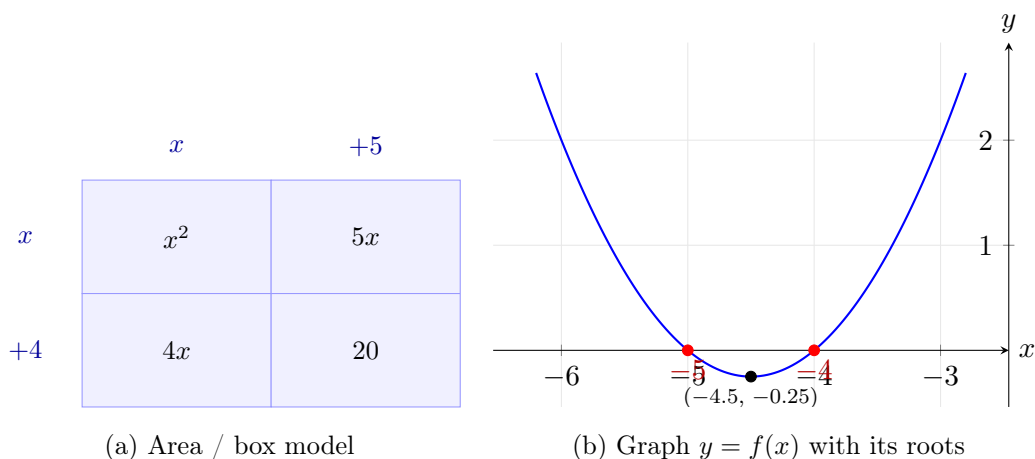
Final answer: $(x + 4)(x + 5)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 5x + 4x + 20$$

Visualising the factorisation



Question 4 qc1-004

Factorise: $x^2 + 11x + 24$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Write down two brackets beginning with x .

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find two numbers multiplying to 24 and adding to 11.

$$3 \times 8 = 24, \quad 3 + 8 = 11$$

Factor pairs of 24: $(1,24), (2,12), (3,8), (4,6)$. Only $(3,8)$ sums to 11.

Step 3. Write the answer.

$$(x + 3)(x + 8)$$

Both signs positive.

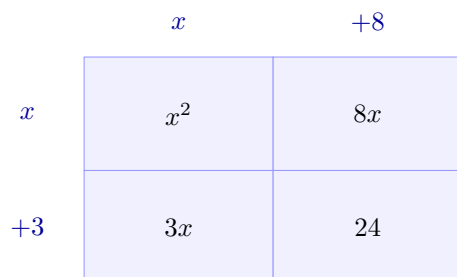
Final answer: $(x + 3)(x + 8)$

Check by expanding

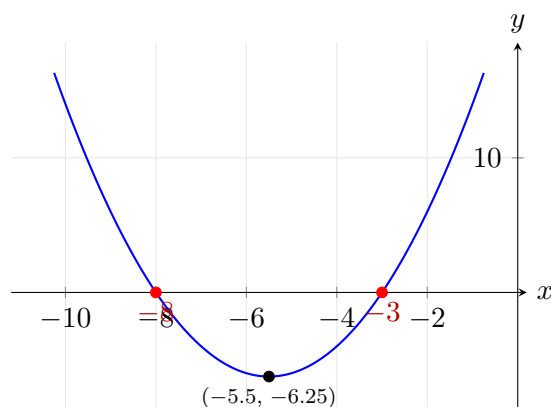
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 8x + 3x + 24$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 5 qc1-005

Factorise: $x^2 + 10x + 21$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up two brackets.

$$(x \quad)(x \quad)$$

Leading coefficient is 1.

Step 2. Find numbers multiplying to 21 and adding to 10.

$$3 \times 7 = 21, \quad 3 + 7 = 10$$

Factor pairs of 21: $(1, 21), (3, 7)$. $(3, 7)$ works.

Step 3. Write the answer.

$$(x + 3)(x + 7)$$

Both signs positive.

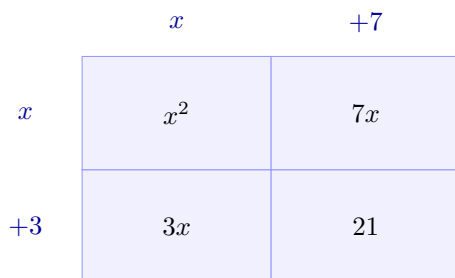
Final answer: $(x + 3)(x + 7)$

Check by expanding

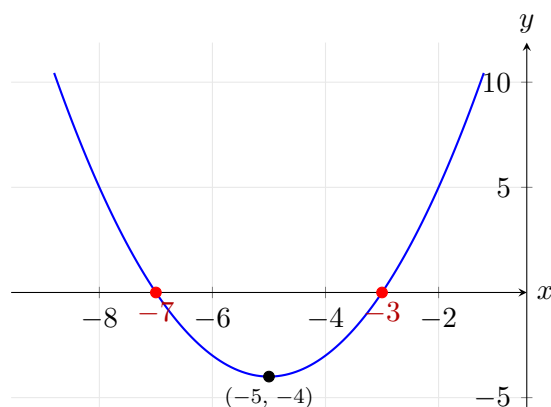
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 7x + 3x + 21$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 6 qc1-006

Factorise: $x^2 - 7x + 10$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find two numbers that multiply to 10 and add to -7 .

$$(-2) \times (-5) = 10, \quad (-2) + (-5) = -7$$

c is positive and b is negative, so both signs are negative.

Step 3. Write the factorised form.

$$(x - 2)(x - 5)$$

Both factors negative.

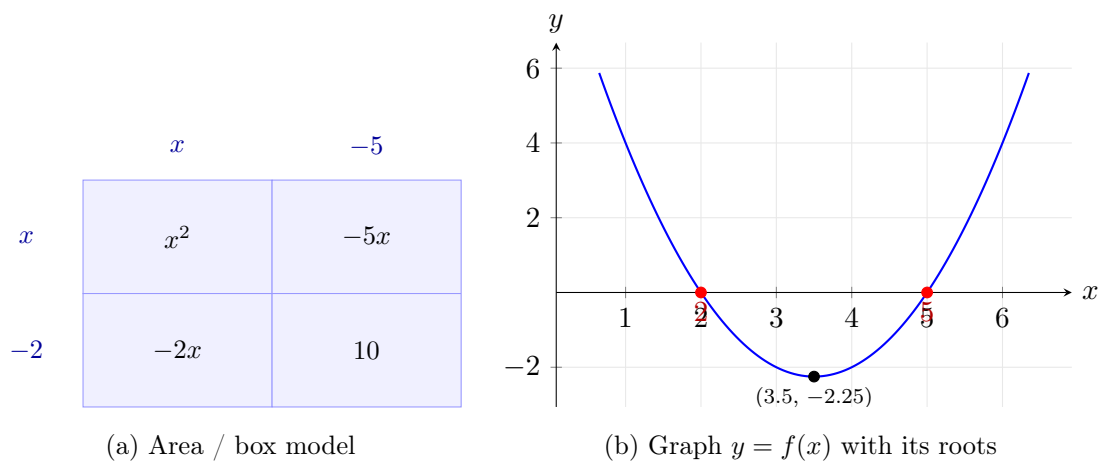
Final answer: $(x - 2)(x - 5)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 5x - 2x + 10$$

Visualising the factorisation



Question 7 qc1-007

Factorise: $x^2 - 9x + 18$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 18 and adding to -9 .

$$(-3) \times (-6) = 18, \quad (-3) + (-6) = -9$$

Both signs negative because $c > 0$ and $b < 0$.

Step 3. Write the answer.

$$(x - 3)(x - 6)$$

Both factors are negative.

Final answer: $(x - 3)(x - 6)$

Check by expanding

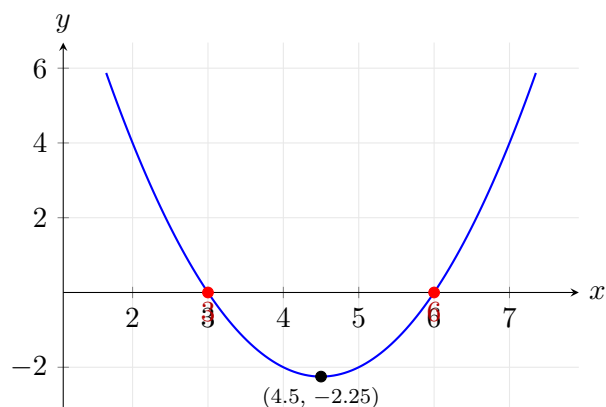
Multiplying the brackets back out reproduces the original expression:

$$x^2 - 6x - 3x + 18$$

Visualising the factorisation

	x	-6
x	x^2	$-6x$
-3	$-3x$	18

(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 8 qc1-008

Factorise: $x^2 - 11x + 28$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Leading coefficient is 1.

Step 2. Find numbers multiplying to 28 and adding to -11 .

$$(-4) \times (-7) = 28, \quad (-4) + (-7) = -11$$

Both signs negative.

Step 3. Write the answer.

$$(x - 4)(x - 7)$$

Both factors are negative.

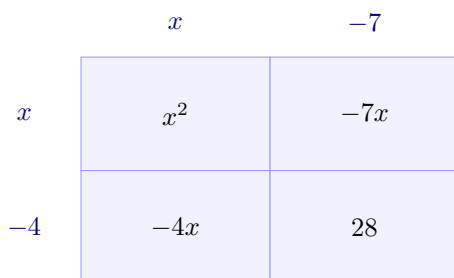
Final answer: $(x - 4)(x - 7)$

Check by expanding

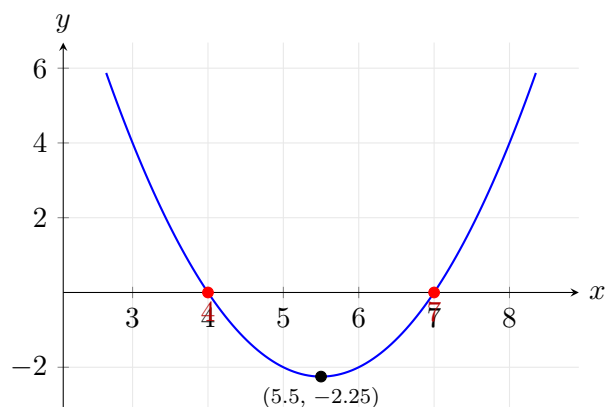
Multiplying the brackets back out reproduces the original expression:

$$x^2 - 7x - 4x + 28$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 9 qc1-009

Factorise: $x^2 - 13x + 36$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 36 and adding to -13 .

$$(-4) \times (-9) = 36, \quad (-4) + (-9) = -13$$

Factor pairs of 36: $(1,36), (2,18), (3,12), (4,9), (6,6)$. The pair $(4,9)$ adds to 13.

Step 3. Write the answer.

$$(x - 4)(x - 9)$$

Both signs negative.

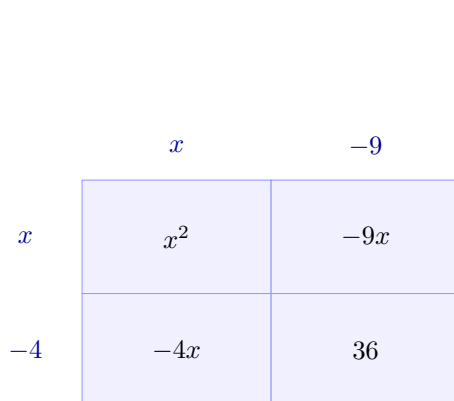
Final answer: $(x - 4)(x - 9)$

Check by expanding

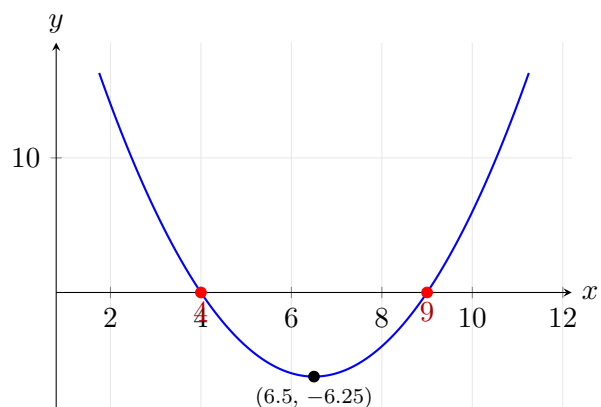
Multiplying the brackets back out reproduces the original expression:

$$x^2 - 9x - 4x + 36$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 10 qc1-010

Factorise: $x^2 - 15x + 56$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 56 and adding to -15 .

$$(-7) \times (-8) = 56, \quad (-7) + (-8) = -15$$

Factor pairs: $(1,56), (2,28), (4,14), (7,8)$. $(7,8)$ adds to 15.

Step 3. Write the answer.

$$(x - 7)(x - 8)$$

Both signs negative.

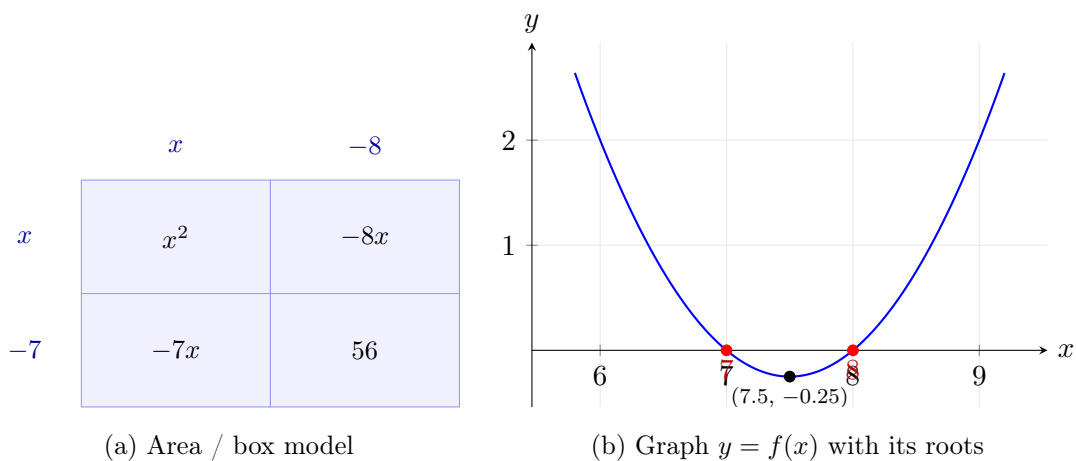
Final answer: $(x - 7)(x - 8)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 8x - 7x + 56$$

Visualising the factorisation



Question 11 qc1-011

Factorise: $x^2 + 2x - 15$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -15 and adding to $+2$.

$$5 \times (-3) = -15, \quad 5 + (-3) = 2$$

c is negative so the signs are opposite.

Step 3. Write the answer.

$$(x + 5)(x - 3)$$

The larger number (5) takes the positive sign.

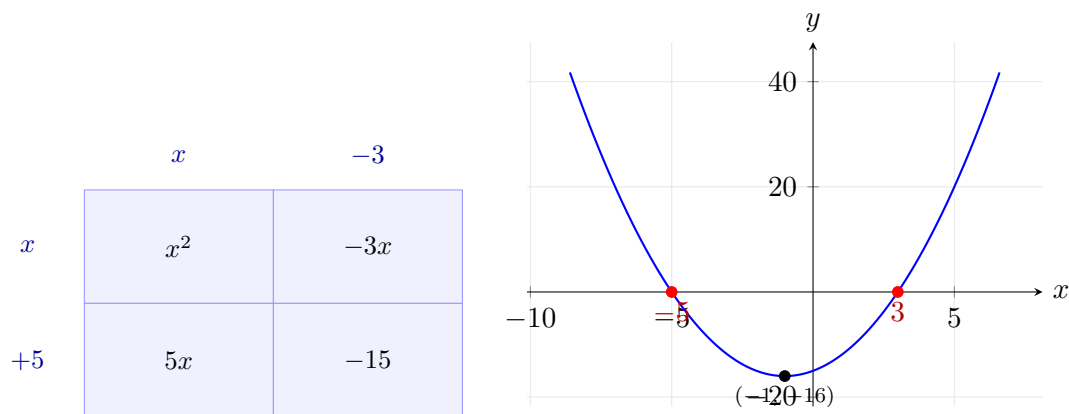
Final answer: $(x + 5)(x - 3)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 3x + 5x - 15$$

Visualising the factorisation



Question 12 qc1-012

Factorise: $x^2 + 4x - 21$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -21 and adding to $+4$.

$$7 \times (-3) = -21, \quad 7 + (-3) = 4$$

Opposite signs because c is negative.

Step 3. Write the answer.

$$(x + 7)(x - 3)$$

The larger value 7 is positive.

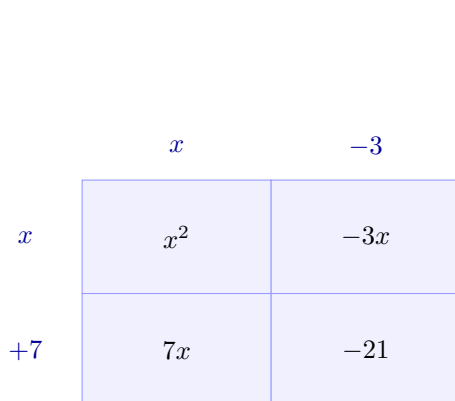
Final answer: $(x + 7)(x - 3)$

Check by expanding

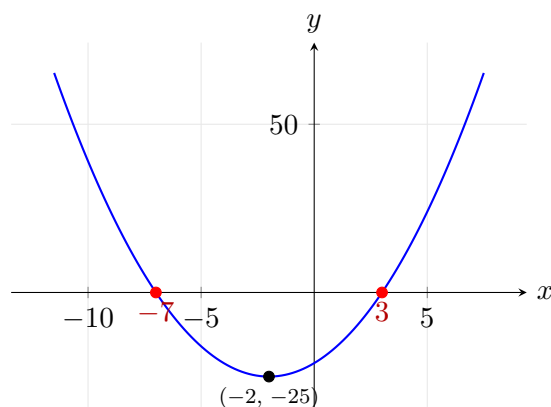
Multiplying the brackets back out reproduces the original expression:

$$x^2 - 3x + 7x - 21$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 13 qc1-013

Factorise: $x^2 + 3x - 28$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -28 and adding to $+3$.

$$7 \times (-4) = -28, \quad 7 + (-4) = 3$$

Factor pairs of 28: $(1,28), (2,14), (4,7)$. $(4,7)$ gives a difference of 3.

Step 3. Write the answer.

$$(x + 7)(x - 4)$$

Larger number (7) is positive.

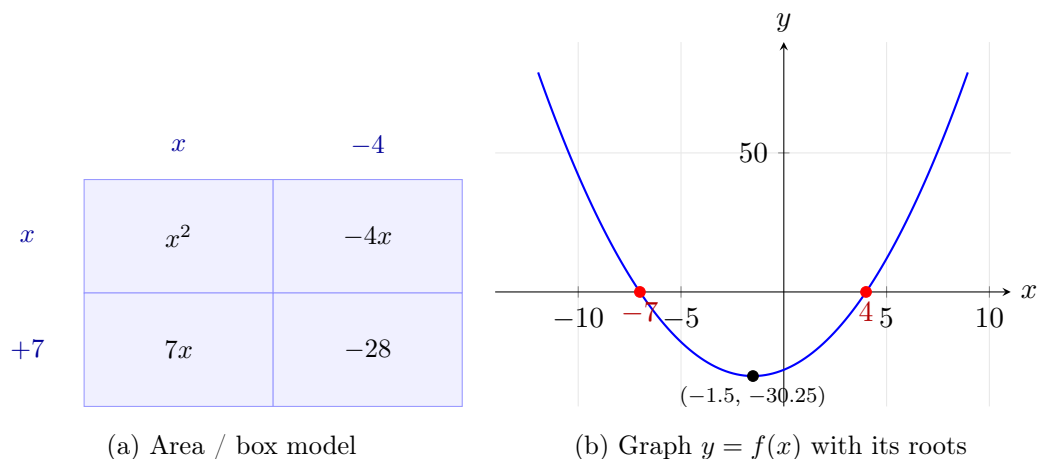
Final answer: $(x + 7)(x - 4)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 4x + 7x - 28$$

Visualising the factorisation



Question 14 qc1-014

Factorise: $x^2 + 5x - 24$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -24 and adding to $+5$.

$$8 \times (-3) = -24, \quad 8 + (-3) = 5$$

Factor pairs of 24: $(1,24), (2,12), (3,8), (4,6)$. $(3,8)$ gives a difference of 5.

Step 3. Write the answer.

$$(x + 8)(x - 3)$$

Larger (8) is positive.

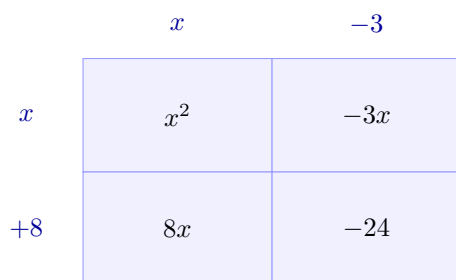
Final answer: $(x + 8)(x - 3)$

Check by expanding

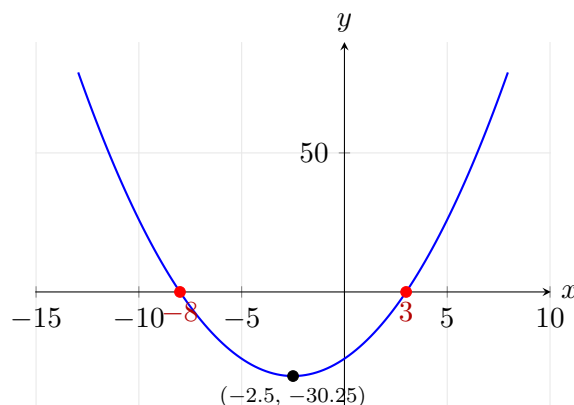
Multiplying the brackets back out reproduces the original expression:

$$x^2 - 3x + 8x - 24$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 15 qc1-015

Factorise: $x^2 + x - 30$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -30 and adding to $+1$.

$$6 \times (-5) = -30, \quad 6 + (-5) = 1$$

Factor pairs of 30: $(1,30), (2,15), (3,10), (5,6)$. $(5,6)$ differs by 1.

Step 3. Write the answer.

$$(x + 6)(x - 5)$$

Larger (6) takes the positive sign.

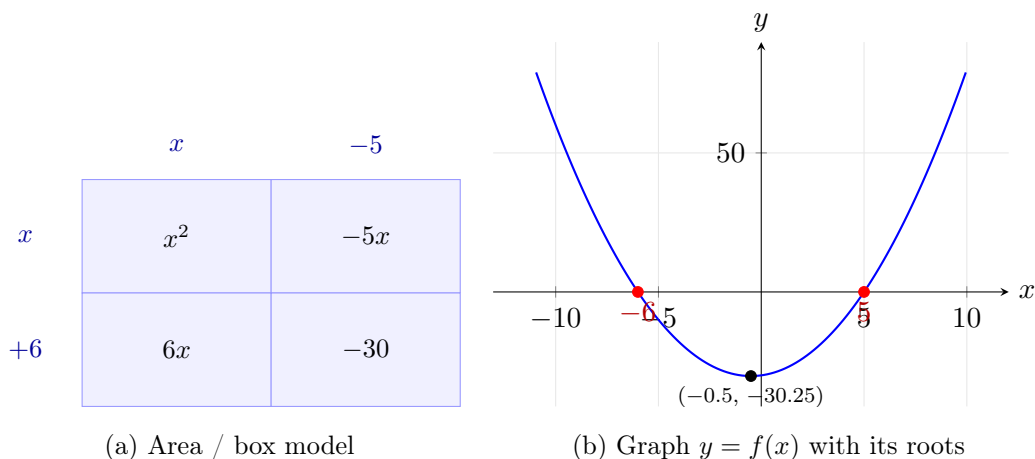
Final answer: $(x + 6)(x - 5)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 5x + 6x - 30$$

Visualising the factorisation



Question 16 qc1-016

Factorise: $x^2 - 2x - 15$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -15 and adding to -2 .

$$(-5) \times 3 = -15, \quad (-5) + 3 = -2$$

Opposite signs; the larger number is negative this time.

Step 3. Write the answer.

$$(x - 5)(x + 3)$$

Larger (5) takes the negative sign.

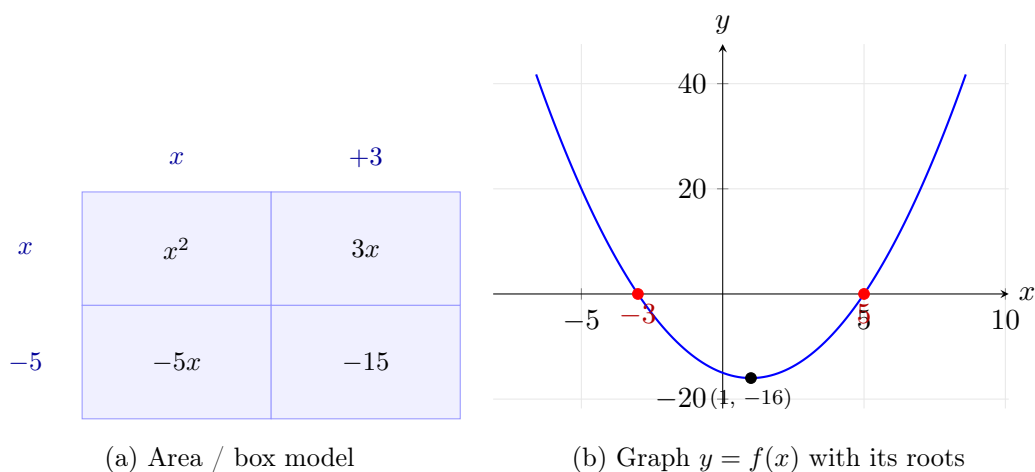
Final answer: $(x - 5)(x + 3)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 3x - 5x - 15$$

Visualising the factorisation



Question 17 qc1-017

Factorise: $x^2 - 3x - 10$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -10 and adding to -3 .

$$(-5) \times 2 = -10, \quad (-5) + 2 = -3$$

Opposite signs; the larger factor (5) is negative.

Step 3. Write the answer.

$$(x - 5)(x + 2)$$

Larger goes with the negative sign.

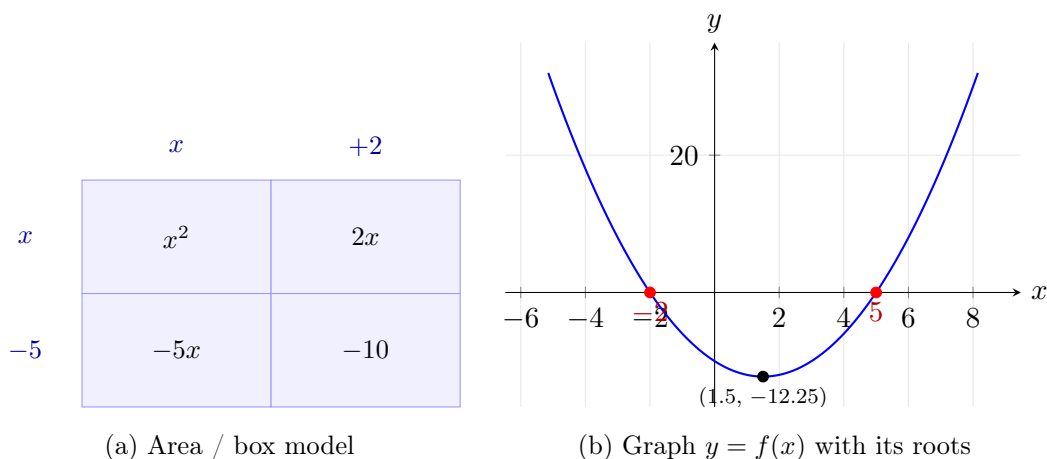
Final answer: $(x - 5)(x + 2)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 2x - 5x - 10$$

Visualising the factorisation



Question 18 qc1-018

Factorise: $x^2 - 5x - 14$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -14 and adding to -5 .

$$(-7) \times 2 = -14, \quad (-7) + 2 = -5$$

Opposite signs with larger factor negative.

Step 3. Write the answer.

$$(x - 7)(x + 2)$$

Larger (7) is negative.

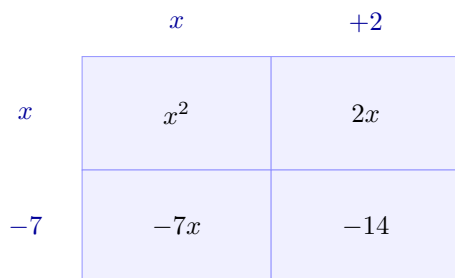
Final answer: $(x - 7)(x + 2)$

Check by expanding

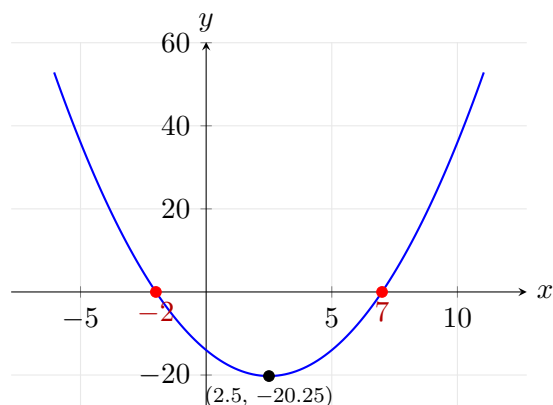
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 2x - 7x - 14$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 19 qc1-019

Factorise: $x^2 - 6x - 16$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -16 and adding to -6 .

$$(-8) \times 2 = -16, \quad (-8) + 2 = -6$$

Opposite signs with larger factor negative.

Step 3. Write the answer.

$$(x - 8)(x + 2)$$

Larger (8) is negative.

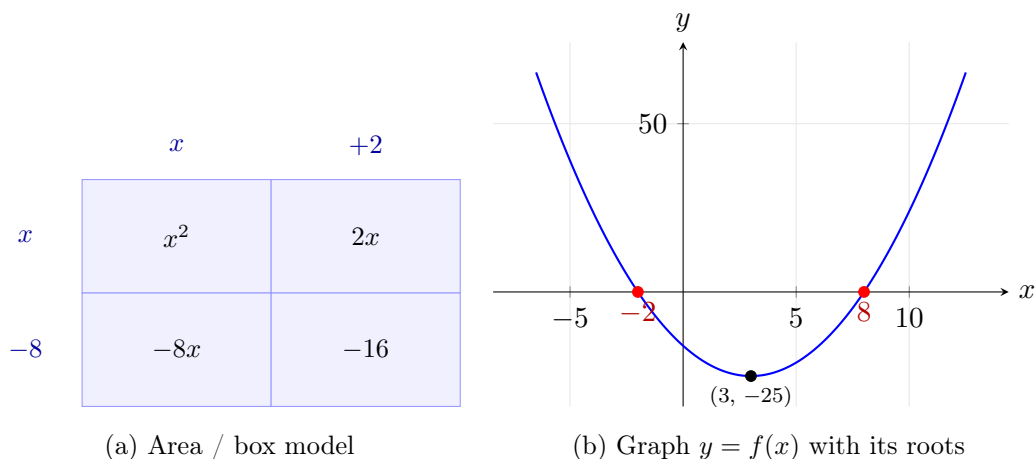
Final answer: $(x - 8)(x + 2)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 2x - 8x - 16$$

Visualising the factorisation



Question 20 qc1-020

Factorise: $x^2 - 4x - 32$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -32 and adding to -4 .

$$(-8) \times 4 = -32, \quad (-8) + 4 = -4$$

Factor pairs of 32: $(1,32), (2,16), (4,8)$. $(4,8)$ differs by 4.

Step 3. Write the answer.

$$(x - 8)(x + 4)$$

Larger (8) is negative.

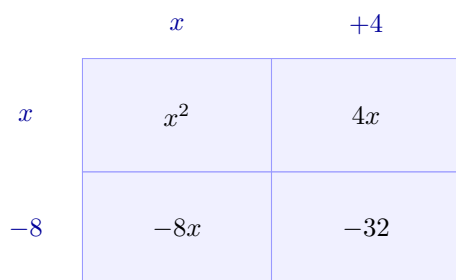
Final answer: $(x - 8)(x + 4)$

Check by expanding

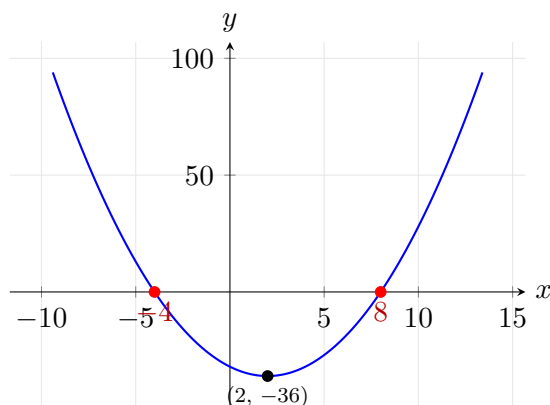
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 4x - 8x - 32$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 21 qc1-021

Factorise: $x^2 + 8x + 16$

Type: Perfect square

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 16 and adding to 8.

$$4 \times 4 = 16, \quad 4 + 4 = 8$$

A repeated factor of 4 works - this is a perfect square.

Step 3. Write the answer.

$$(x + 4)(x + 4) = (x + 4)^2$$

Repeated factor gives a perfect square.

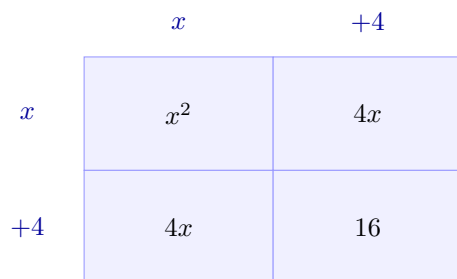
Final answer: $(x + 4)^2$

Check by expanding

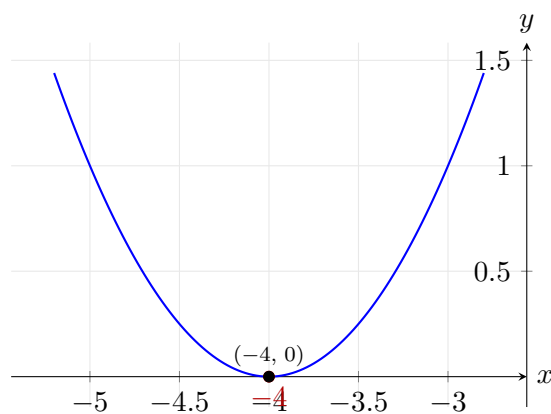
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 4x + 4x + 16$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 22 qc1-022

Factorise: $x^2 - 10x + 25$

Type: Perfect square

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 25 and adding to -10 .

$$(-5) \times (-5) = 25, \quad (-5) + (-5) = -10$$

Repeated factor of -5 .

Step 3. Write the answer.

$$(x - 5)^2$$

Perfect square.

Final answer: $(x - 5)^2$

Check by expanding

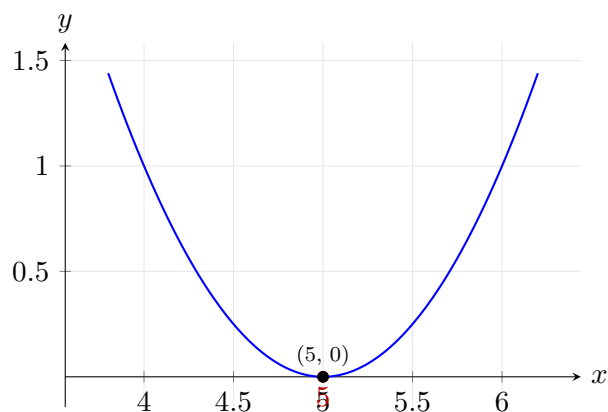
Multiplying the brackets back out reproduces the original expression:

$$x^2 - 5x - 5x + 25$$

Visualising the factorisation

	x	-5
x	x^2	$-5x$
-5	$-5x$	25

(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 23 qc1-023

Factorise: $x^2 + 12x + 36$

Type: Perfect square

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 36 and adding to 12.

$$6 \times 6 = 36, \quad 6 + 6 = 12$$

Repeated factor of 6.

Step 3. Write the answer.

$$(x + 6)^2$$

Perfect square.

Final answer: $(x + 6)^2$

Check by expanding

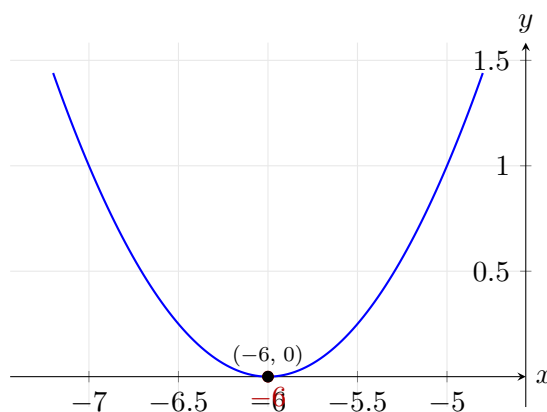
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 6x + 6x + 36$$

Visualising the factorisation

	x	$+6$
x	x^2	$6x$
$+6$	$6x$	36

(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 24 qc1-024

Factorise: $x^2 - 14x + 49$

Type: Perfect square

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 49 and adding to -14 .

$$(-7) \times (-7) = 49, \quad (-7) + (-7) = -14$$

Repeated factor of -7 .

Step 3. Write the answer.

$$(x - 7)^2$$

Perfect square.

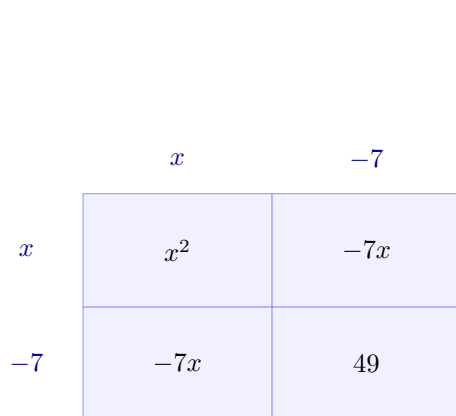
Final answer: $(x - 7)^2$

Check by expanding

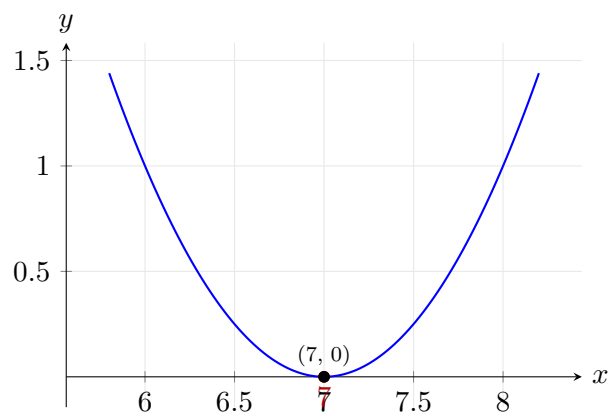
Multiplying the brackets back out reproduces the original expression:

$$x^2 - 7x - 7x + 49$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 25 qc1-025

Factorise: $x^2 - 25$

Type: Difference of two squares

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Recognise the difference of two squares.

$$x^2 - 25 = x^2 - 5^2$$

Both x^2 and 25 are perfect squares separated by a minus sign.

Step 2. Apply $a^2 - b^2 = (a - b)(a + b)$.

$$(x - 5)(x + 5)$$

Replace a with x and b with 5.

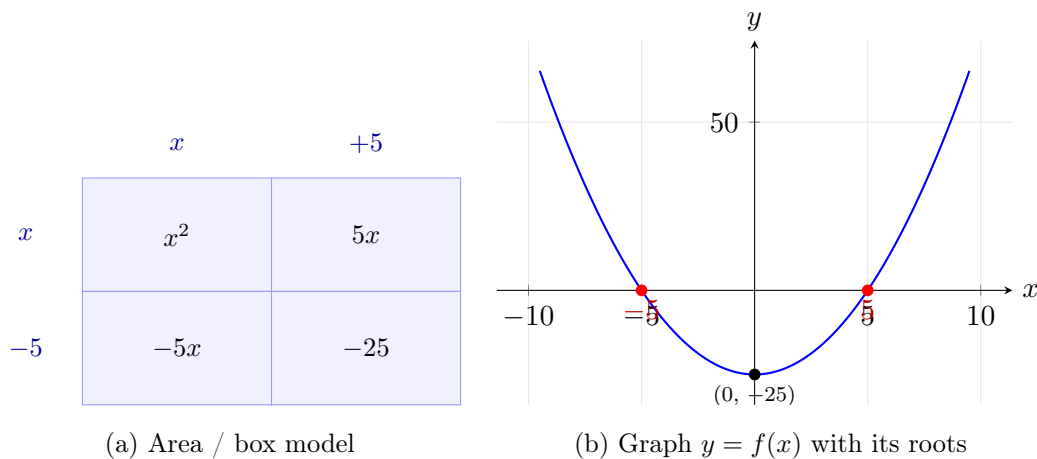
Final answer: $(x - 5)(x + 5)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 5x - 5x - 25$$

Visualising the factorisation



Question 26 qc1-026

Factorise: $x^2 - 64$

Type: Difference of two squares

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Recognise the difference of two squares.

$$x^2 - 64 = x^2 - 8^2$$

$$64 = 8^2.$$

Step 2. Apply the difference of two squares formula.

$$(x - 8)(x + 8)$$

One bracket with a minus, one with a plus.

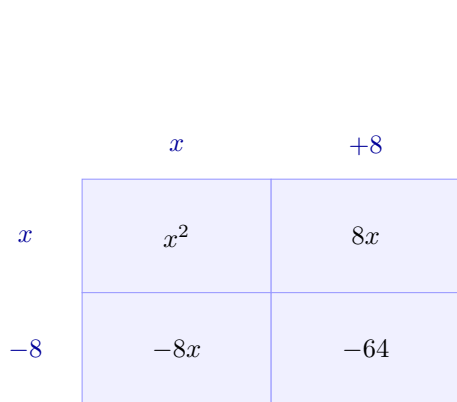
Final answer: $(x - 8)(x + 8)$

Check by expanding

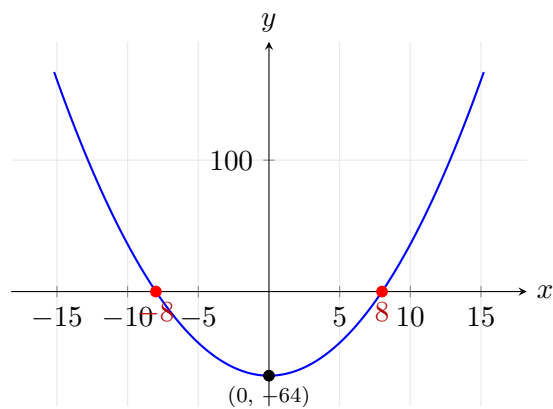
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 8x - 8x - 64$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 27 qc1-027

Factorise: $x^2 - 100$

Type: Difference of two squares

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Recognise the difference of two squares.

$$x^2 - 100 = x^2 - 10^2$$

$$100 = 10^2.$$

Step 2. Apply the formula.

$$(x - 10)(x + 10)$$

Difference of two squares.

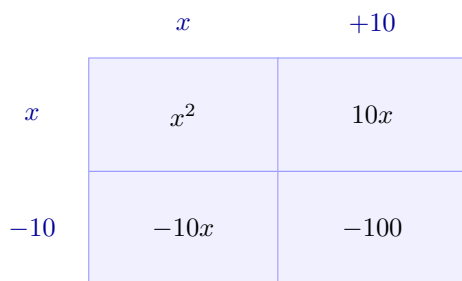
Final answer: $(x - 10)(x + 10)$

Check by expanding

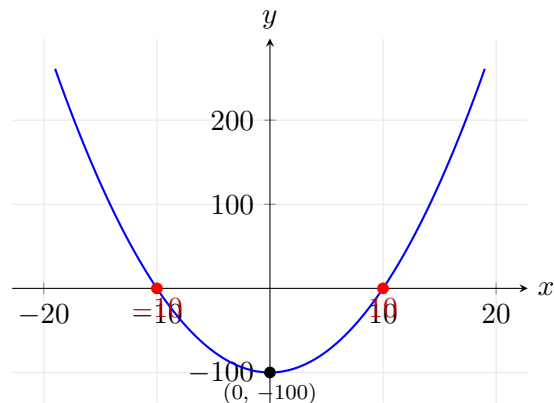
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 10x - 10x - 100$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 28 qc1-028

Factorise: $x^2 - 81$

Type: Difference of two squares

Marks: 2 *Difficulty:* Foundation

Worked solution

Step 1. Recognise the difference of two squares.

$$x^2 - 81 = x^2 - 9^2$$

$$81 = 9^2.$$

Step 2. Apply the formula.

$$(x - 9)(x + 9)$$

Difference of two squares.

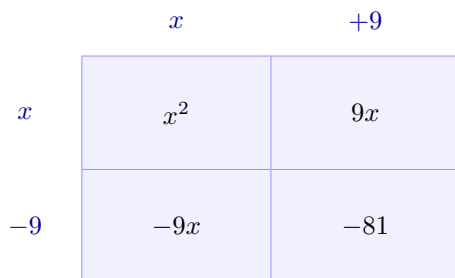
Final answer: $(x - 9)(x + 9)$

Check by expanding

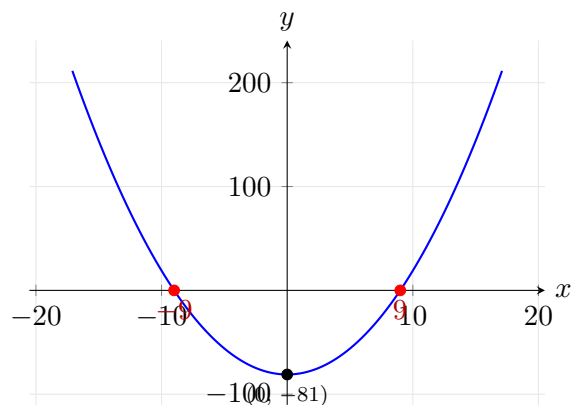
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 9x - 9x - 81$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 29

qc1-029

Factorise: $x^2 - 8x$

Type: Common factor

Marks: 2 *Difficulty:* Foundation

Worked solution

Step 1. Find the common factor in both terms.

$$x^2 - 8x$$

Both terms contain an x .

Step 2. Factor out x .

$$x(x - 8)$$

Dividing each term by x gives x and -8 .

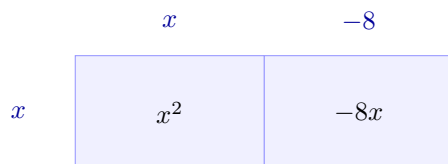
Final answer: $x(x - 8)$

Check by expanding

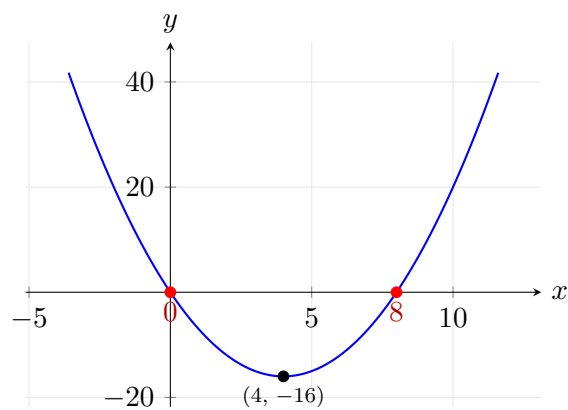
Multiplying the brackets back out reproduces the original expression:

$$x^2 - 8x$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 30 qc1-030

Factorise: $x^2 + 11x$

Type: Common factor

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Identify the common factor.

$$x^2 + 11x$$

Both terms share an x .

Step 2. Factor out x .

$$x(x + 11)$$

Pulling out x leaves $x + 11$ inside the bracket.

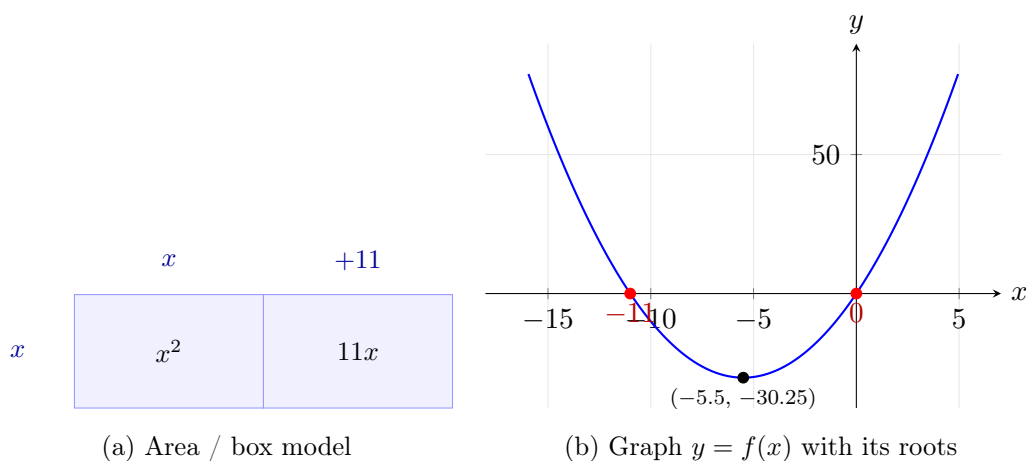
Final answer: $x(x + 11)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 11x$$

Visualising the factorisation



Question 31 qc1-031

Factorise: $2x^2 + 7x + 3$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets that will give $2x^2$ when multiplied.

$$(2x \quad)(x \quad)$$

$$2x \times x = 2x^2.$$

Step 2. Find numbers multiplying to 3 that combine with $2x$ and x to give $7x$.

$$(2x + 1)(x + 3) : 2x \cdot 3 + 1 \cdot x = 6x + x = 7x$$

Try the pair (1, 3) and check which arrangement gives $7x$.

Step 3. Write the answer.

$$(2x + 1)(x + 3)$$

All signs positive.

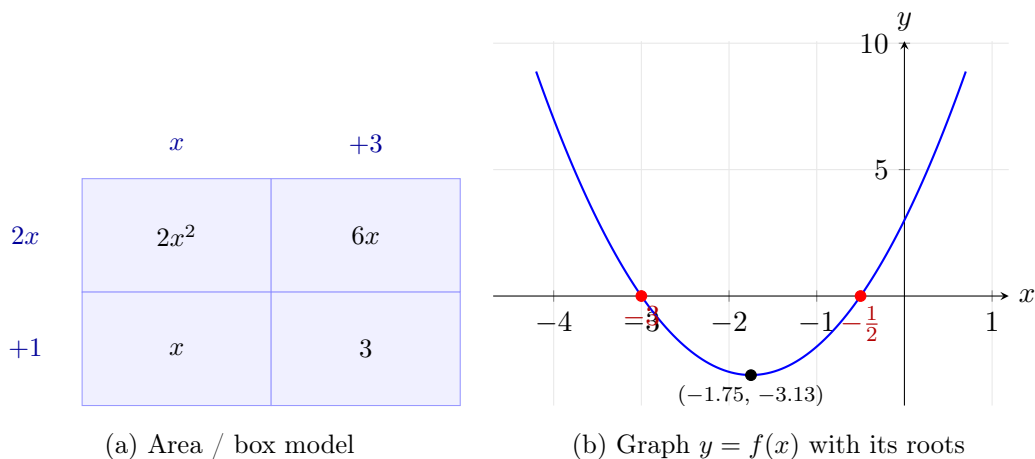
Final answer: $(2x + 1)(x + 3)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$2x^2 + 6x + x + 3$$

Visualising the factorisation



Question 32 qc1-032

Factorise: $2x^2 + 9x + 4$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(2x \quad)(x \quad)$$

$$2x \times x = 2x^2.$$

Step 2. Try pairs that multiply to 4 and give a middle term of 9x.

$$(2x + 1)(x + 4) : 2x \cdot 4 + 1 \cdot x = 8x + x = 9x$$

The pair (1, 4) works with 1 in the first bracket.

Step 3. Write the answer.

$$(2x + 1)(x + 4)$$

All signs positive.

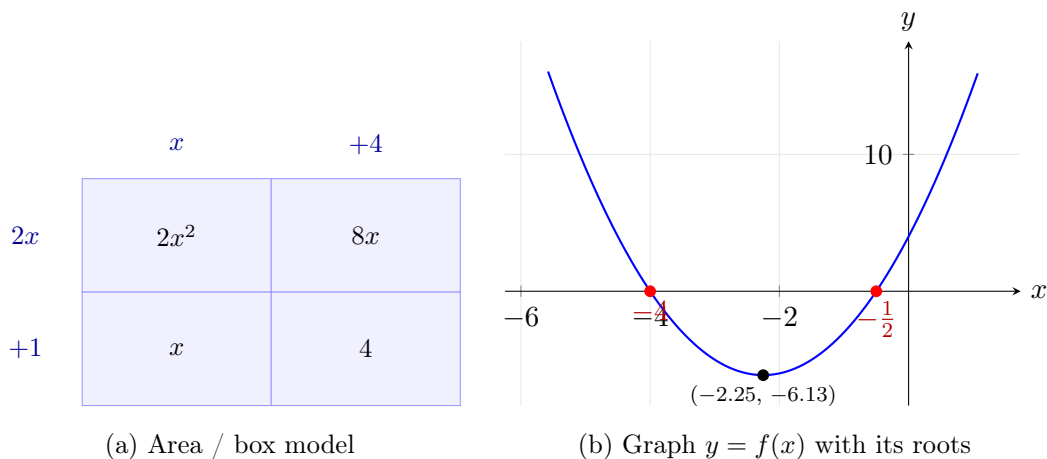
Final answer: $(2x + 1)(x + 4)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$2x^2 + 8x + x + 4$$

Visualising the factorisation



Question 33 qc1-033

Factorise: $2x^2 + 11x + 5$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(2x \quad)(x \quad)$$

$$2x \times x = 2x^2.$$

Step 2. Find a pair multiplying to 5 that gives middle term $11x$.

$$(2x + 1)(x + 5) : 10x + x = 11x$$

Put 1 with $2x$ and 5 with x .

Step 3. Write the answer.

$$(2x + 1)(x + 5)$$

All signs positive.

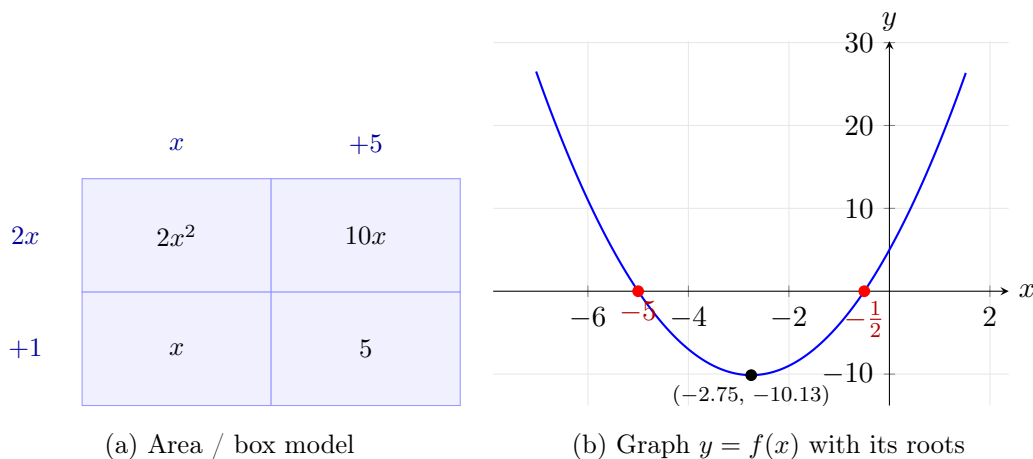
Final answer: $(2x + 1)(x + 5)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$2x^2 + 10x + x + 5$$

Visualising the factorisation



Question 34 qc1-034

Factorise: $3x^2 + 10x + 3$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(3x \quad)(x \quad)$$

$$3x \times x = 3x^2.$$

Step 2. Find a pair multiplying to 3 that gives middle term $10x$.

$$(3x + 1)(x + 3) : 9x + x = 10x$$

Put 1 with $3x$ and 3 with x .

Step 3. Write the answer.

$$(3x + 1)(x + 3)$$

All signs positive.

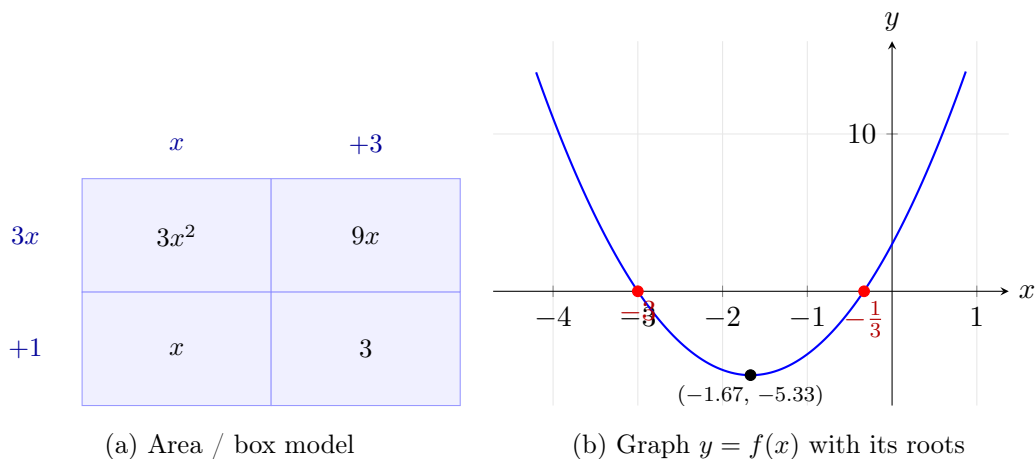
Final answer: $(3x + 1)(x + 3)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$3x^2 + 9x + x + 3$$

Visualising the factorisation



Question 35 qc1-035

Factorise: $3x^2 + 8x + 4$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(3x \quad)(x \quad)$$

$$3x \times x = 3x^2.$$

Step 2. Find a pair multiplying to 4 that gives middle term $8x$.

$$(3x + 2)(x + 2) : 6x + 2x = 8x$$

Pair (2, 2) works.

Step 3. Write the answer.

$$(3x + 2)(x + 2)$$

All signs positive.

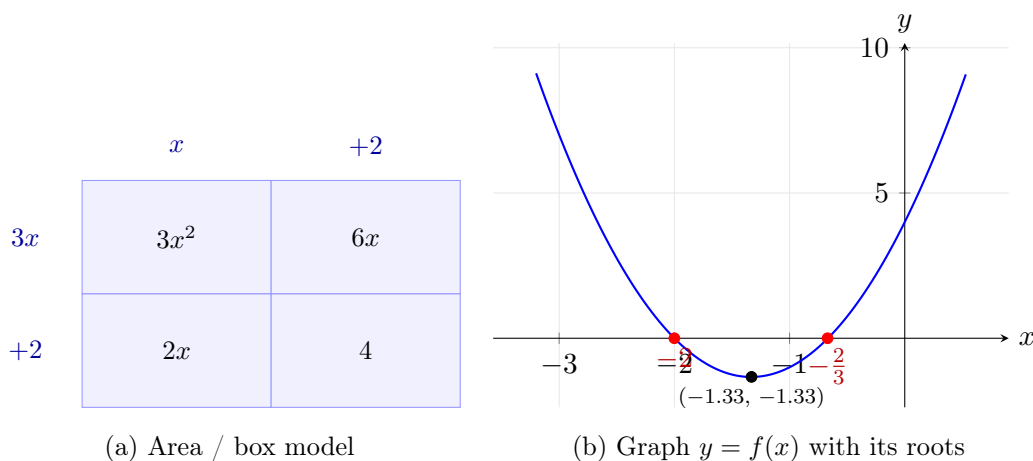
Final answer: $(3x + 2)(x + 2)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$3x^2 + 6x + 2x + 4$$

Visualising the factorisation



Question 36 qc1-036

Factorise: $2x^2 - 7x + 3$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(2x \quad)(x \quad)$$

$$2x \times x = 2x^2.$$

Step 2. Since $c > 0$ and $b < 0$, both signs are negative. Find a pair with product 3 giving middle term $-7x$.

$$(2x - 1)(x - 3) : -6x - x = -7x$$

Place -1 with 2x and -3 with x.

Step 3. Write the answer.

$$(2x - 1)(x - 3)$$

Both signs negative.

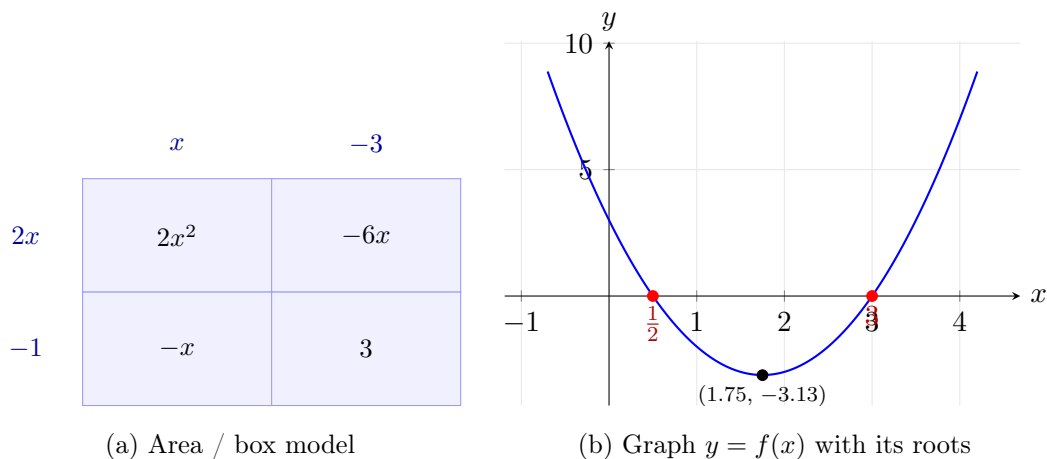
Final answer: $(2x - 1)(x - 3)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$2x^2 - 6x - x + 3$$

Visualising the factorisation



Question 37 qc1-037

Factorise: $3x^2 - 11x + 6$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(3x \quad)(x \quad)$$

$$3x \times x = 3x^2.$$

Step 2. Both signs negative. Try pairs of 6 that give $-11x$ in the middle.

$$(3x - 2)(x - 3) : -9x - 2x = -11x$$

-2 with $3x$ and -3 with x .

Step 3. Write the answer.

$$(3x - 2)(x - 3)$$

Both signs negative.

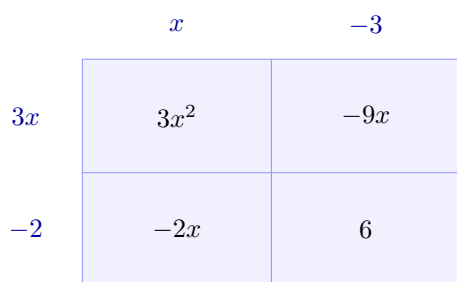
Final answer: $(3x - 2)(x - 3)$

Check by expanding

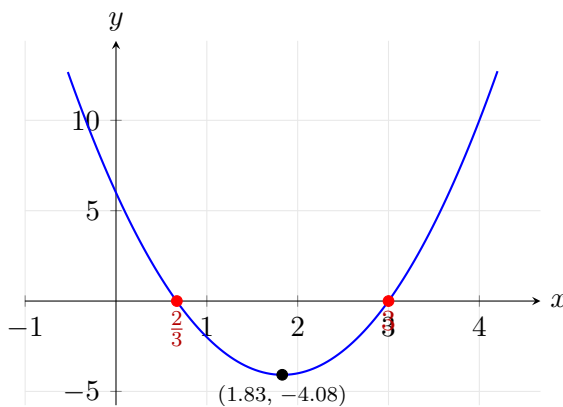
Multiplying the brackets back out reproduces the original expression:

$$3x^2 - 9x - 2x + 6$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 38 qc1-038

Factorise: $2x^2 + 5x - 3$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(2x \quad)(x \quad)$$

$$2x \times x = 2x^2.$$

Step 2. Since $c < 0$, the signs are opposite. Find a combination giving $+5x$.

$$(2x - 1)(x + 3) : 6x - x = 5x$$

Place -1 with $2x$ and $+3$ with x .

Step 3. Write the answer.

$$(2x - 1)(x + 3)$$

Opposite signs.

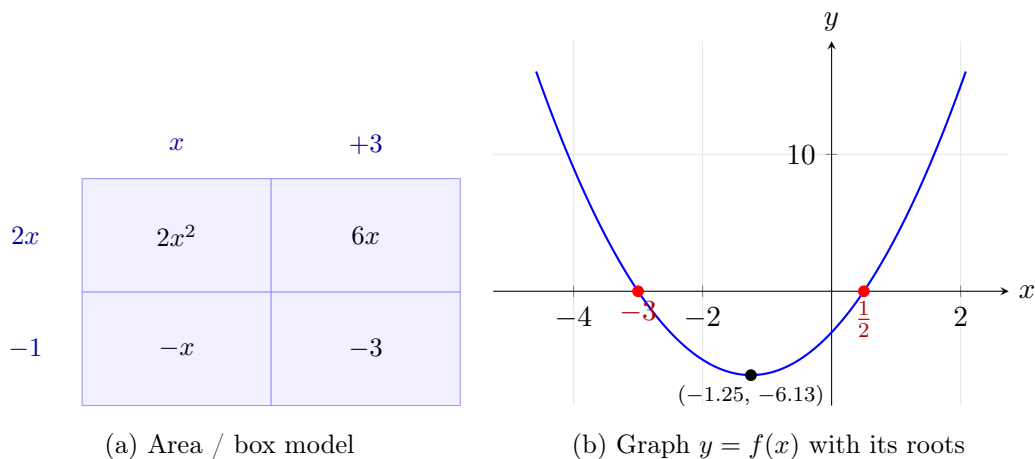
Final answer: $(2x - 1)(x + 3)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$2x^2 + 6x - x - 3$$

Visualising the factorisation



Question 39 qc1-039

Factorise: $3x^2 + 5x - 2$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(3x \quad)(x \quad)$$

$$3x \times x = 3x^2.$$

Step 2. c is negative so signs are opposite. Find a combination giving $+5x$.

$$(3x - 1)(x + 2) : 6x - x = 5x$$

-1 with $3x$ and $+2$ with x .

Step 3. Write the answer.

$$(3x - 1)(x + 2)$$

Opposite signs.

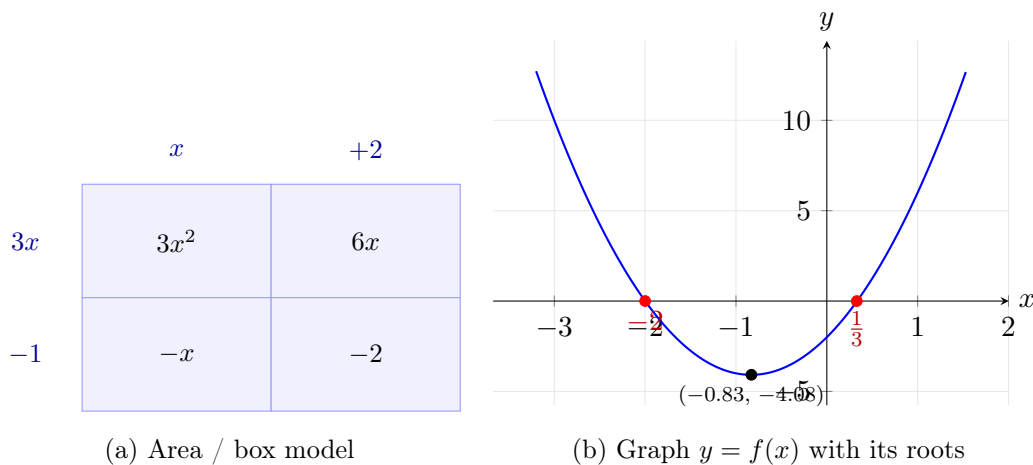
Final answer: $(3x - 1)(x + 2)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$3x^2 + 6x - x - 2$$

Visualising the factorisation



Question 40 qc1-040

Factorise: $2x^2 - 5x - 3$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(2x \quad)(x \quad)$$

$$2x \times x = 2x^2.$$

Step 2. Opposite signs needed. Find a combination giving $-5x$.

$$(2x + 1)(x - 3) : -6x + x = -5x$$

$+1$ with $2x$ and -3 with x .

Step 3. Write the answer.

$$(2x + 1)(x - 3)$$

Opposite signs.

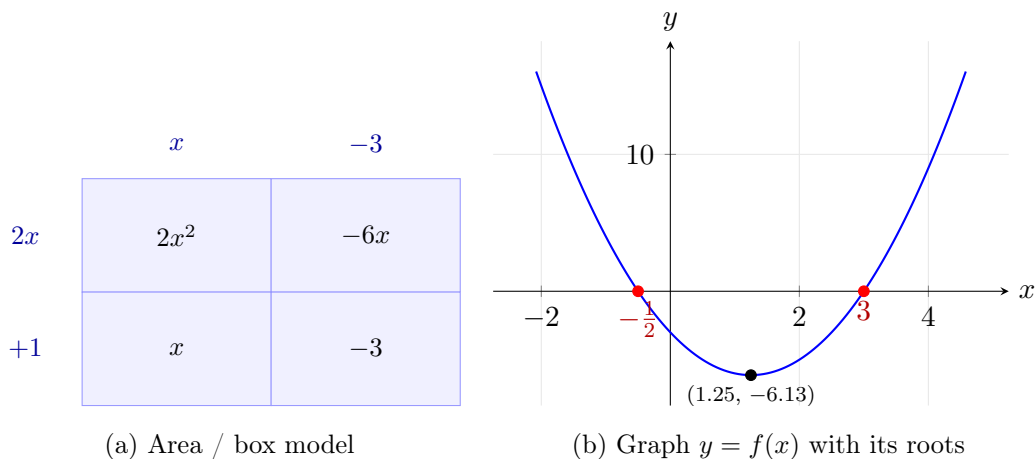
Final answer: $(2x + 1)(x - 3)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$2x^2 - 6x + x - 3$$

Visualising the factorisation



Question 41 qc1-041

Factorise: $3x^2 - 7x - 6$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(3x \quad)(x \quad)$$

$$3x \times x = 3x^2.$$

Step 2. Opposite signs. Try $(3x + 2)(x - 3)$.

$$(3x + 2)(x - 3) : -9x + 2x = -7x$$

+2 with $3x$ and -3 with x gives the correct middle term.

Step 3. Write the answer.

$$(3x + 2)(x - 3)$$

Opposite signs.

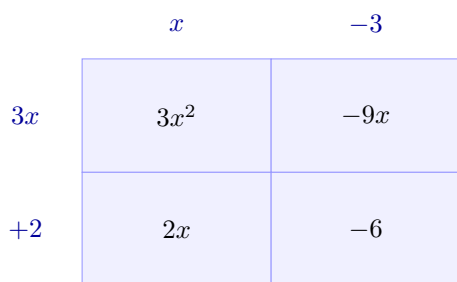
Final answer: $(3x + 2)(x - 3)$

Check by expanding

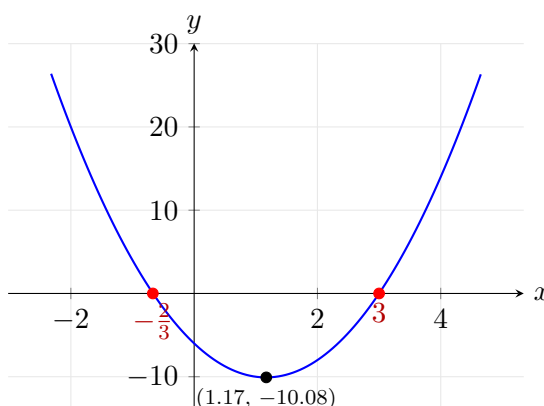
Multiplying the brackets back out reproduces the original expression:

$$3x^2 - 9x + 2x - 6$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 42 qc1-042

Factorise: $4x^2 - 9$

Type: Difference of two squares

Marks: 2 *Difficulty:* Foundation

Worked solution

Step 1. Recognise the difference of two squares.

$$4x^2 - 9 = (2x)^2 - 3^2$$

Both $4x^2$ and 9 are perfect squares.

Step 2. Apply $a^2 - b^2 = (a - b)(a + b)$.

$$(2x - 3)(2x + 3)$$

Here $a = 2x$ and $b = 3$.

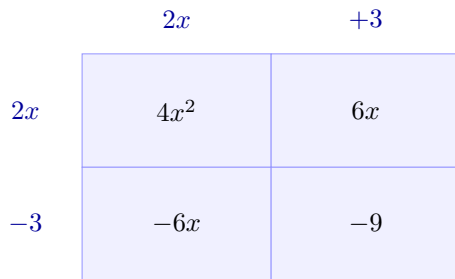
Final answer: $(2x - 3)(2x + 3)$

Check by expanding

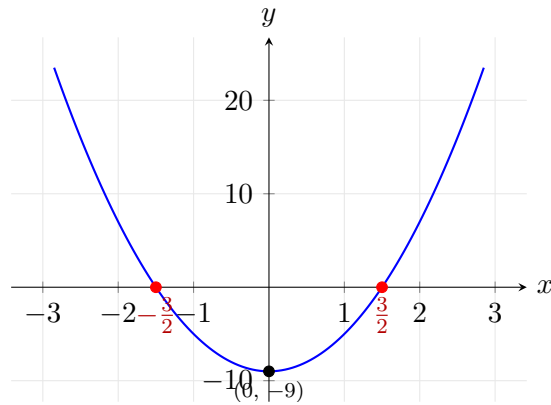
Multiplying the brackets back out reproduces the original expression:

$$4x^2 + 6x - 6x - 9$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 43

qc1-043

Factorise: $9x^2 - 16$

Type: Difference of two squares

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Recognise the difference of two squares.

$$9x^2 - 16 = (3x)^2 - 4^2$$

$$9x^2 = (3x)^2 \text{ and } 16 = 4^2.$$

Step 2. Apply the formula.

$$(3x - 4)(3x + 4)$$

Here $a = 3x$ and $b = 4$.

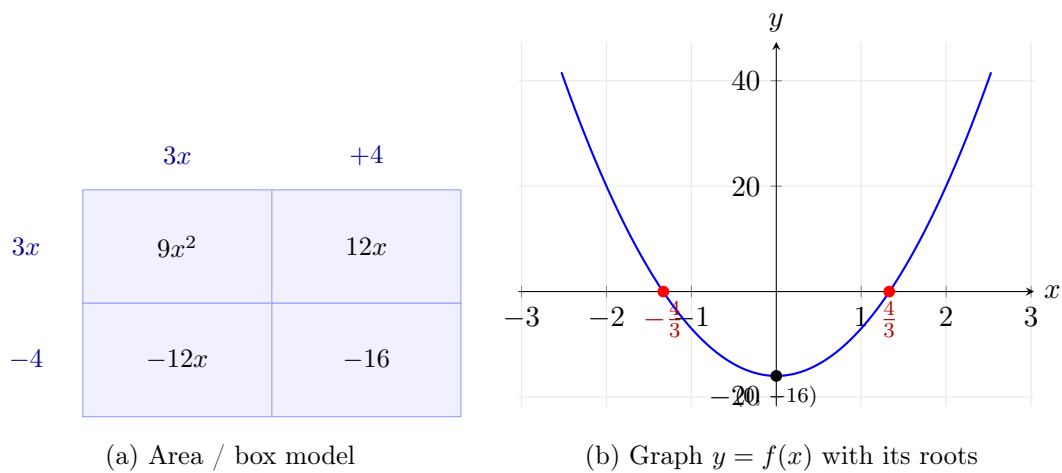
Final answer: $(3x - 4)(3x + 4)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$9x^2 + 12x - 12x - 16$$

Visualising the factorisation



Question 44 qc1-044

Factorise: $25x^2 - 49$

Type: Difference of two squares

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Recognise the difference of two squares.

$$25x^2 - 49 = (5x)^2 - 7^2$$

$$25x^2 = (5x)^2 \text{ and } 49 = 7^2.$$

Step 2. Apply the formula.

$$(5x - 7)(5x + 7)$$

Here $a = 5x$ and $b = 7$.

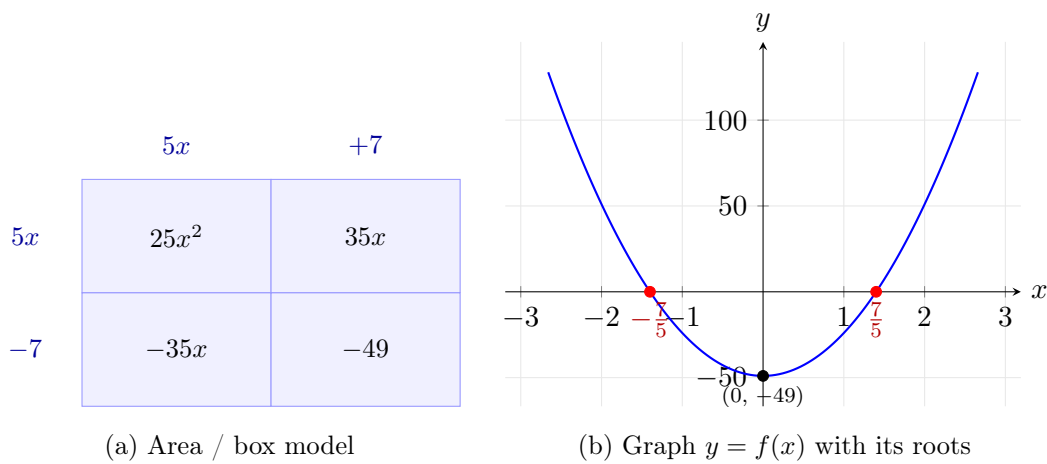
Final answer: $(5x - 7)(5x + 7)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$25x^2 + 35x - 35x - 49$$

Visualising the factorisation



Question 45 qc1-045

Factorise: $2x^2 - 18$

Type: Difference of two squares

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Take out the common factor 2.

$$2(x^2 - 9)$$

Both terms share a factor of 2.

Step 2. Factorise $x^2 - 9$ as a difference of two squares.

$$2(x - 3)(x + 3)$$

$$x^2 - 9 = (x - 3)(x + 3).$$

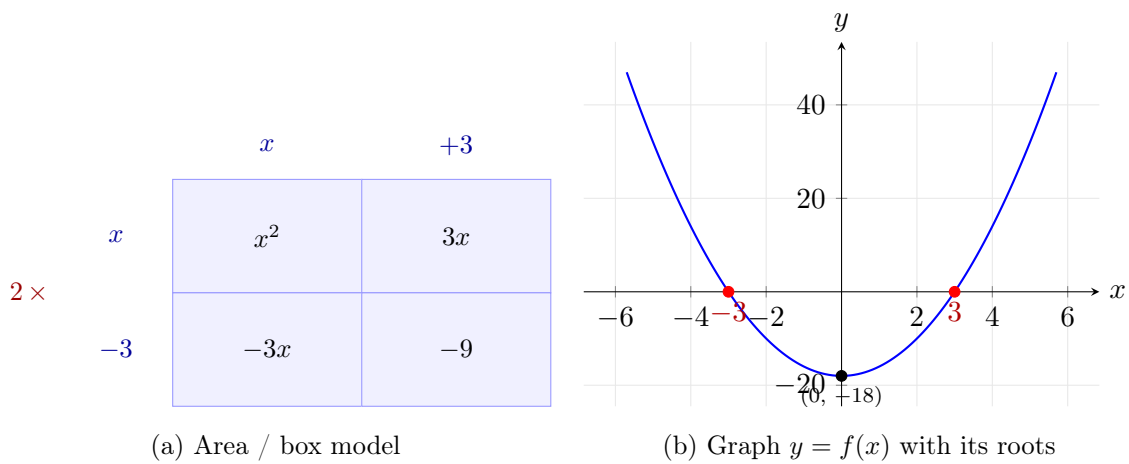
Final answer: $2(x - 3)(x + 3)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$2(x^2 + 3x - 3x - 9)$$

Visualising the factorisation



Question 46 qc1-046

Factorise: $3x^2 + 12x + 9$

Type: Common factor

Marks: 3 Difficulty: Foundation

Worked solution

Step 1. Take out the common factor 3.

$$3(x^2 + 4x + 3)$$

All three terms share a factor of 3.

Step 2. Factorise the bracket: numbers multiplying to 3 and adding to 4.

$$1 \times 3 = 3, \quad 1 + 3 = 4$$

The pair (1, 3) works.

Step 3. Write the answer.

$$3(x + 1)(x + 3)$$

Keep the factor 3 outside.

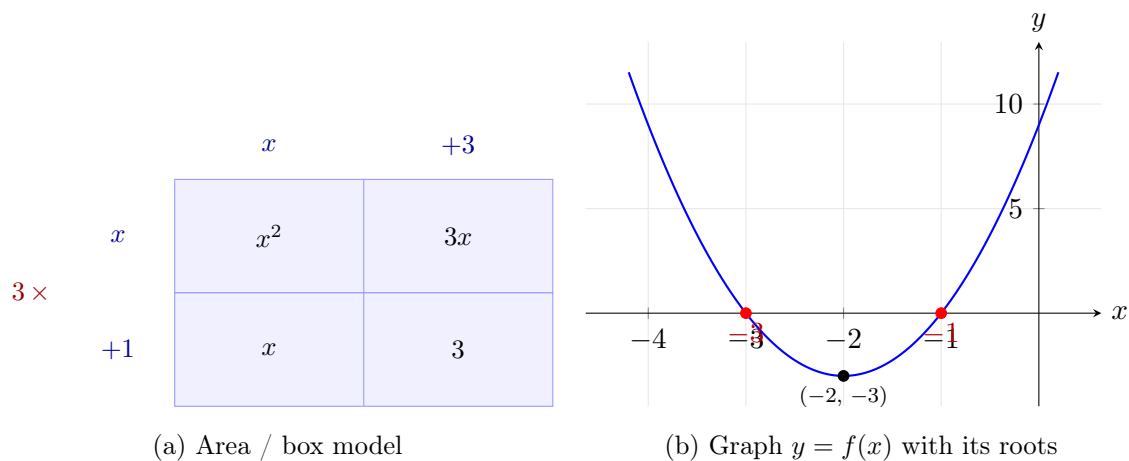
Final answer: $3(x + 1)(x + 3)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$3(x^2 + 3x + x + 3)$$

Visualising the factorisation



Question 47 qc1-047

Solve by factorising: $x^2 + 6x + 8 = 0$

Type: Solving by factorising

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Factorise the quadratic.

$$(x + 2)(x + 4) = 0$$

2 and 4 multiply to 8 and add to 6.

Step 2. Set each factor equal to zero.

$$x + 2 = 0 \text{ or } x + 4 = 0$$

If a product is zero, at least one factor must be zero.

Step 3. Solve each equation.

$$x = -2 \text{ or } x = -4$$

These are the two roots.

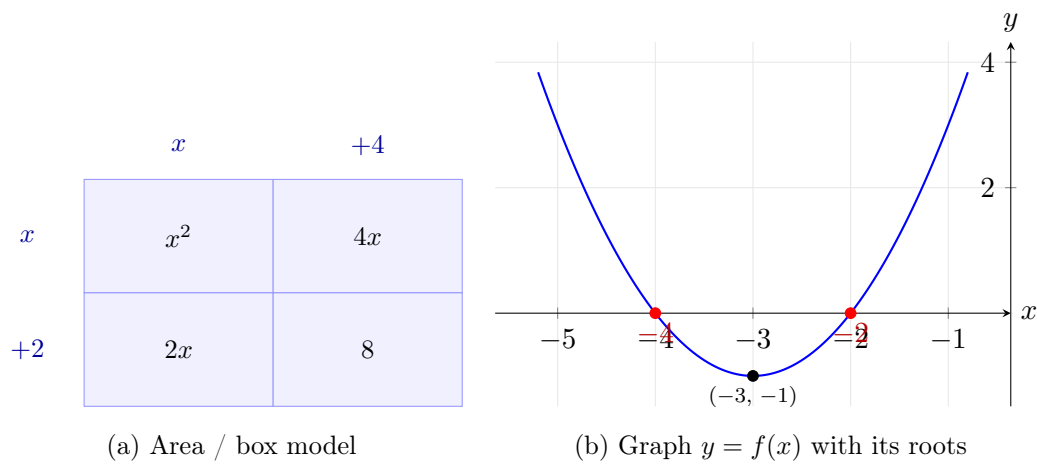
Final answer: $x = -2$ or $x = -4$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 4x + 2x + 8$$

Visualising the factorisation



Question 48 qc1-048

Solve by factorising: $x^2 - 5x + 6 = 0$

Type: Solving by factorising

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Factorise the quadratic.

$$(x - 2)(x - 3) = 0$$

$$(-2) \times (-3) = 6 \text{ and } -2 - 3 = -5.$$

Step 2. Set each factor equal to zero.

$$x - 2 = 0 \text{ or } x - 3 = 0$$

One of the factors must be zero.

Step 3. Solve for x .

$$x = 2 \text{ or } x = 3$$

These are the two solutions.

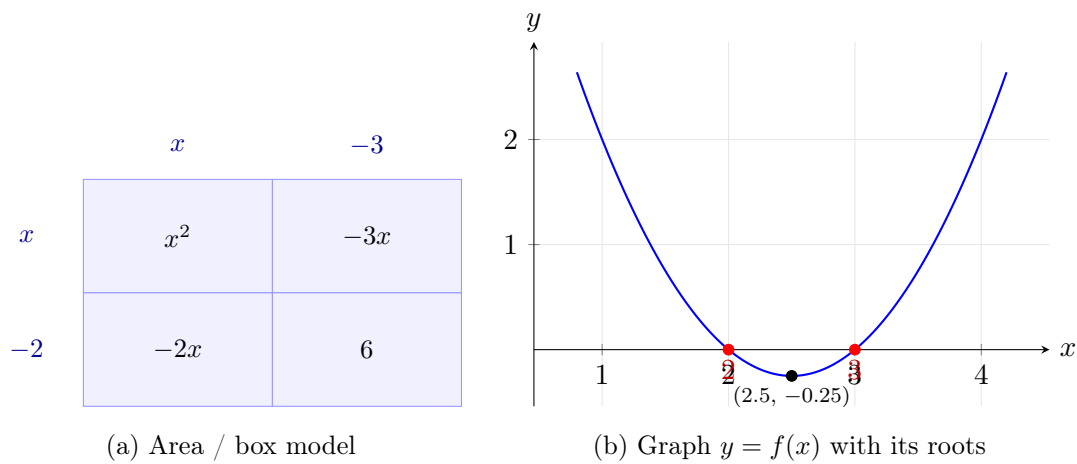
Final answer: $x = 2$ or $x = 3$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 3x - 2x + 6$$

Visualising the factorisation



Question 49 qc1-049

Solve by factorising: $x^2 - x - 12 = 0$

Type: Solving by factorising

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Factorise the quadratic.

$$(x - 4)(x + 3) = 0$$

$$(-4) \times 3 = -12 \text{ and } -4 + 3 = -1.$$

Step 2. Set each factor equal to zero.

$$x - 4 = 0 \text{ or } x + 3 = 0$$

One factor must be zero.

Step 3. Solve for x .

$$x = 4 \text{ or } x = -3$$

The two solutions.

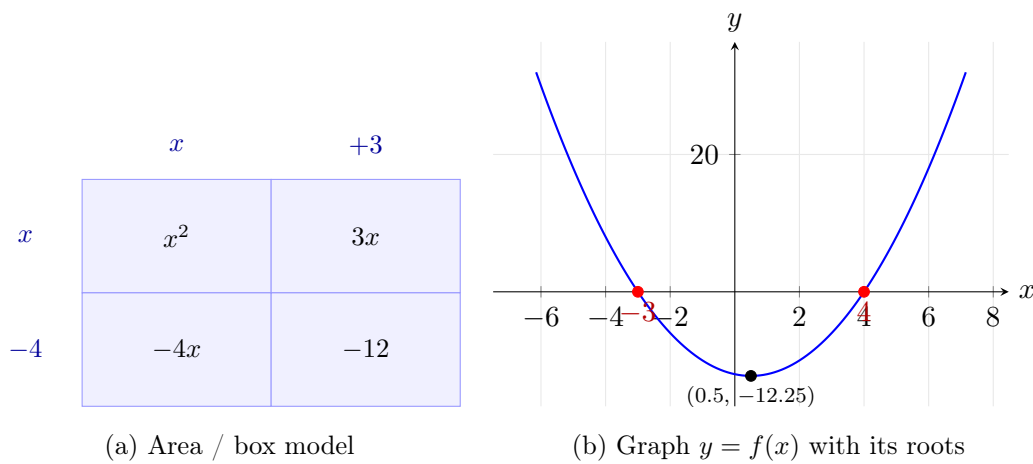
Final answer: $x = 4$ or $x = -3$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 3x - 4x - 12$$

Visualising the factorisation



Question 50 qc1-050

Solve by factorising: $x^2 - 9 = 0$

Type: Difference of two squares

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Recognise the difference of two squares and factorise.

$$(x - 3)(x + 3) = 0$$

$$x^2 - 9 = x^2 - 3^2.$$

Step 2. Set each factor equal to zero.

$$x - 3 = 0 \text{ or } x + 3 = 0$$

One of the factors must be zero.

Step 3. Solve for x .

$$x = 3 \text{ or } x = -3$$

Both roots are ± 3 .

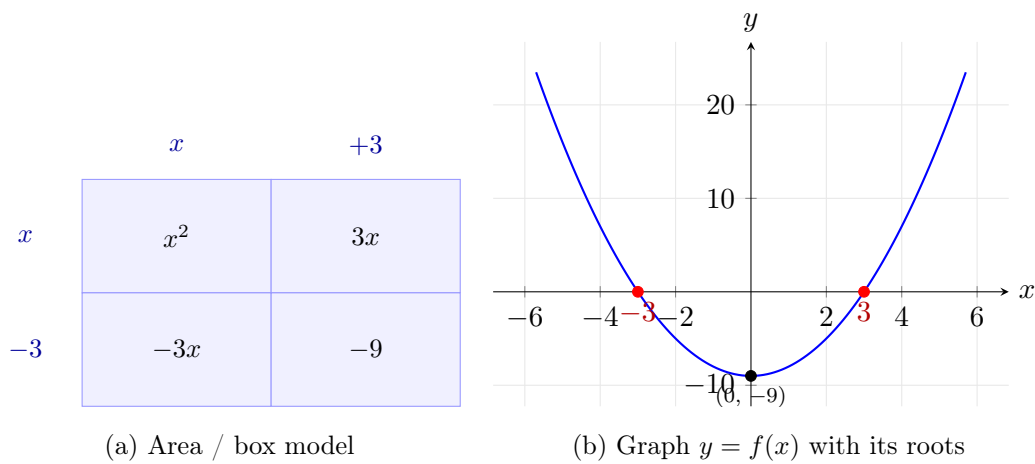
Final answer: $x = 3$ or $x = -3$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 3x - 3x - 9$$

Visualising the factorisation



Question 51 qc1-051

Solve by factorising: $x^2 - 2x - 8 = 0$

Type: Solving by factorising

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Factorise the quadratic. Find two numbers that multiply to -8 and add to -2 .

$$(x - 4)(x + 2) = 0$$

$$-4 \times 2 = -8 \text{ and } -4 + 2 = -2.$$

Step 2. Set each factor equal to zero.

$$x - 4 = 0 \text{ or } x + 2 = 0$$

If a product is zero, at least one factor is zero.

Step 3. Solve each equation.

$$x = 4 \text{ or } x = -2$$

These are the two roots.

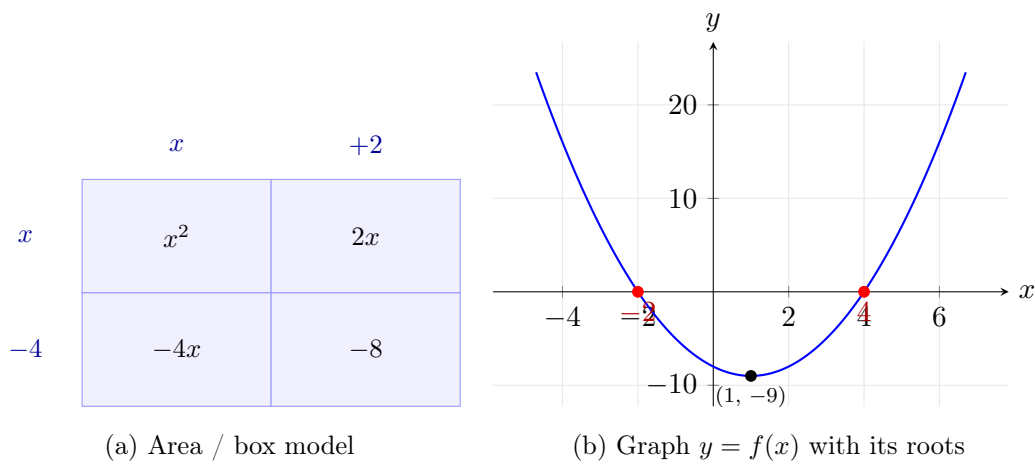
Final answer: $x = 4$ or $x = -2$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 2x - 4x - 8$$

Visualising the factorisation



Question 52 qc1-052

Solve by factorising: $2x^2 + 2x - 40 = 0$

Type: Common factor

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Divide through by 2 to simplify first.

$$x^2 + x - 20 = 0$$

All three coefficients share the factor 2.

Step 2. Factorise: find numbers multiplying to -20 and adding to $+1$.

$$(x + 5)(x - 4) = 0$$

$$5 \times (-4) = -20 \text{ and } 5 + (-4) = 1.$$

Step 3. Set each factor to zero and solve.

$$x = -5 \text{ or } x = 4$$

Two solutions from the product-is-zero rule.

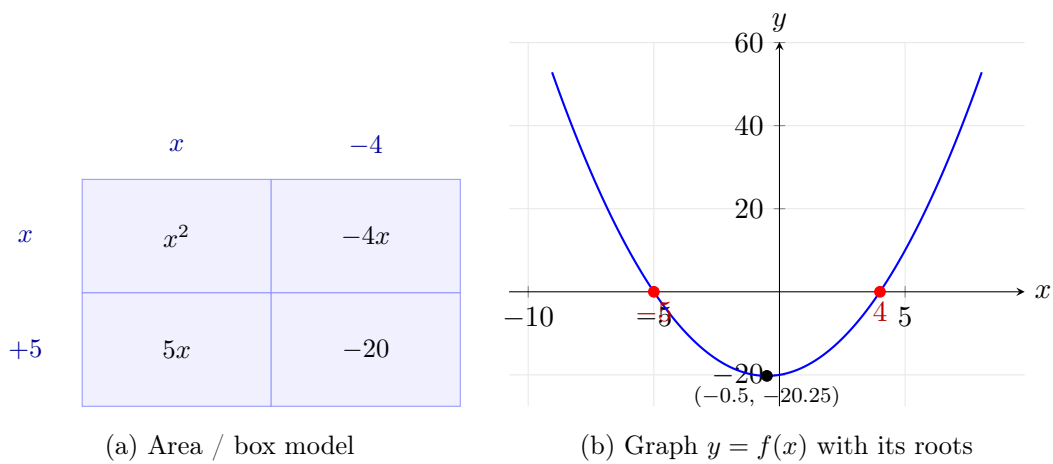
Final answer: $x = -5$ or $x = 4$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 4x + 5x - 20$$

Visualising the factorisation



Question 53 qc1-053

Solve by factorising: $p^2 + 21p + 38 = 0$

Type: Solving by factorising

Marks: 3 Difficulty: Foundation

Worked solution

Step 1. Find two numbers multiplying to 38 and adding to 21.

$$2 \times 19 = 38, \quad 2 + 19 = 21$$

Factor pairs of 38 are (1,38) and (2,19). Only (2,19) sums to 21.

Step 2. Write the factorised equation.

$$(p + 2)(p + 19) = 0$$

Both signs positive.

Step 3. Set each factor to zero and solve.

$$p = -2 \text{ or } p = -19$$

The two solutions.

Final answer: $p = -2$ or $p = -19$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$1 + 19 + 2 + 38$$

Visualising the factorisation

	1	+19
1	1	19
+2	2	38

Area / box model: each cell is the product of its row and column term.

Question 54 qc1-054

Solve by factorising: $x^2 - 15x + 54 = 0$

Type: Solving by factorising

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Find numbers multiplying to 54 and adding to -15 .

$$(-6) \times (-9) = 54, \quad (-6) + (-9) = -15$$

Factor pairs of 54: $(1, 54), (2, 27), (3, 18), (6, 9)$. Only $(6, 9)$ sums to 15.

Step 2. Factorise and set to zero.

$$(x - 6)(x - 9) = 0$$

Both signs negative.

Step 3. Solve.

$$x = 6 \text{ or } x = 9$$

The two solutions.

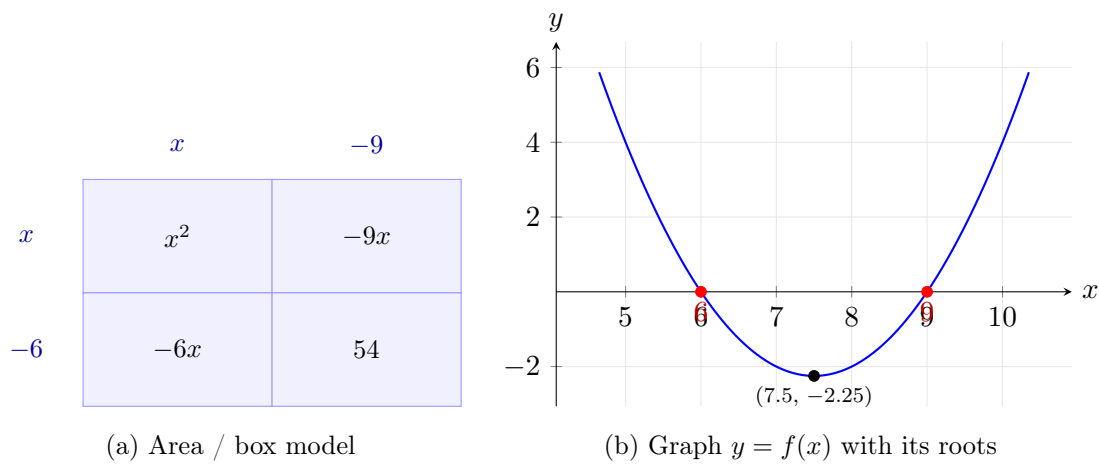
Final answer: $x = 6$ or $x = 9$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 9x - 6x + 54$$

Visualising the factorisation



Question 55 qc1-055

Solve by factorising: $x^2 + 18x = -65$

Type: Solving by factorising

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Rearrange into standard form.

$$x^2 + 18x + 65 = 0$$

Add 65 to both sides.

Step 2. Find numbers multiplying to 65 and adding to 18.

$$5 \times 13 = 65, \quad 5 + 13 = 18$$

Factor pairs of 65: (1,65), (5,13). (5,13) works.

Step 3. Factorise and solve.

$$(x + 5)(x + 13) = 0 \Rightarrow x = -5 \text{ or } x = -13$$

Set each factor to zero.

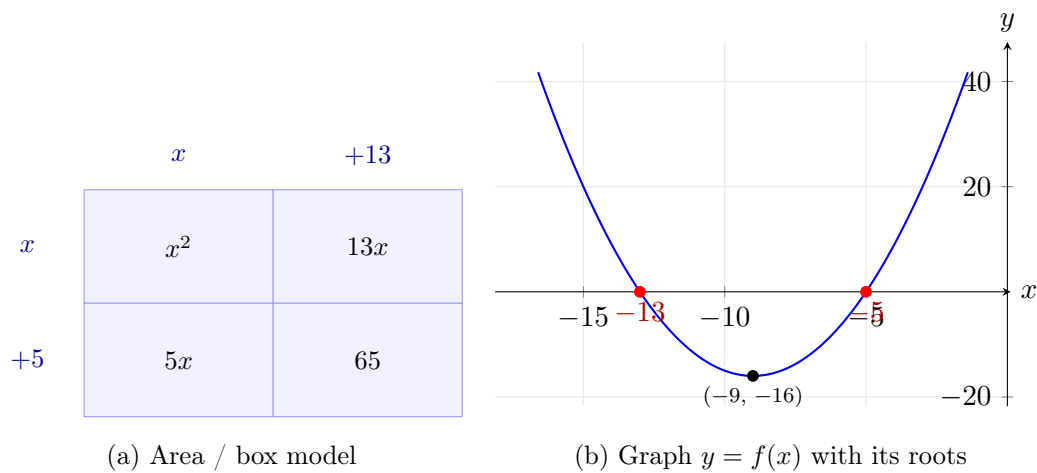
Final answer: $x = -5$ or $x = -13$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 13x + 5x + 65$$

Visualising the factorisation



Question 56 qc1-056

Solve by factorising: $x^2 - x = 42$

Type: Solving by factorising

Marks: 3 Difficulty: Foundation

Worked solution

Step 1. Rearrange into standard form.

$$x^2 - x - 42 = 0$$

Subtract 42 from both sides.

Step 2. Find numbers multiplying to -42 and adding to -1 .

$$(-7) \times 6 = -42, \quad (-7) + 6 = -1$$

Opposite signs; larger factor (7) is negative.

Step 3. Factorise and solve.

$$(x - 7)(x + 6) = 0 \Rightarrow x = 7 \text{ or } x = -6$$

Set each factor to zero.

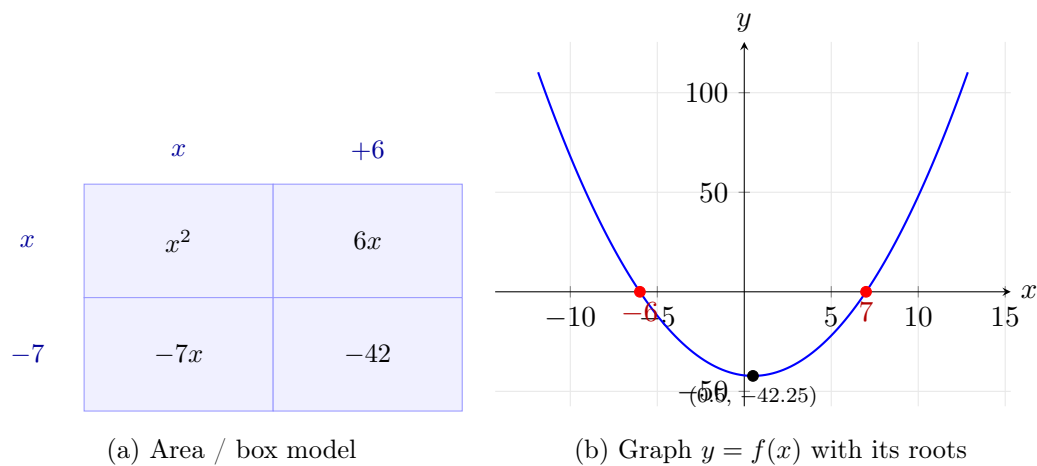
Final answer: $x = 7$ or $x = -6$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 6x - 7x - 42$$

Visualising the factorisation



Question 57 qc1-057

Solve by factorising: $3x^2 - 3x - 6 = 0$

Type: Common factor

Marks: 3 Difficulty: Foundation

Worked solution

Step 1. Divide through by 3.

$$x^2 - x - 2 = 0$$

All terms share a factor of 3.

Step 2. Factorise: numbers multiplying to -2 , adding to -1 .

$$(x - 2)(x + 1) = 0$$

$$-2 \times 1 = -2 \text{ and } -2 + 1 = -1.$$

Step 3. Solve.

$$x = 2 \text{ or } x = -1$$

The two roots.

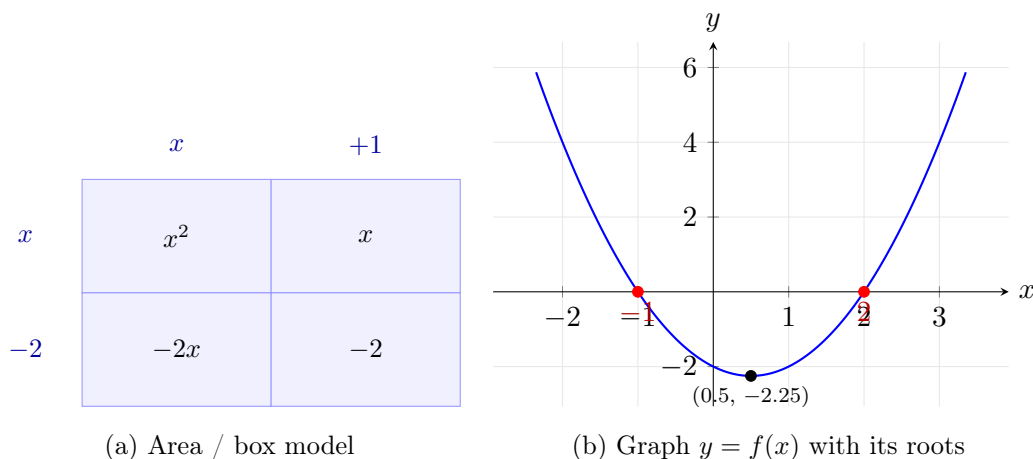
Final answer: $x = 2$ or $x = -1$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + x - 2x - 2$$

Visualising the factorisation



Question 58 qc1-058

Solve by factorising: $5x^2 - 21x + 4 = 0$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets for the non-monic case.

$$(5x \quad)(x \quad) = 0$$

$$5x \times x = 5x^2.$$

Step 2. Both signs negative ($c > 0$, $b < 0$). Try pairs of 4.

$$(5x - 1)(x - 4) : -20x - x = -21x$$

-1 with $5x$ and -4 with x produces $-21x$.

Step 3. Solve each factor.

$$5x - 1 = 0 \text{ or } x - 4 = 0 \Rightarrow x = \frac{1}{5} \text{ or } x = 4$$

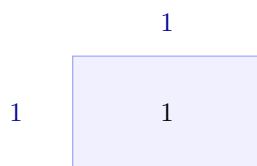
Divide by 5 for the first root.

Final answer: $x = \frac{1}{5}$ or $x = 4$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

Visualising the factorisation



Area / box model: each cell is the product of its row and column term.

Question 59 qc1-059

Solve by factorising: $2x^2 + 11x + 5 = 0$

Type: Non-monic ($a \neq 1$)

Marks: 3 *Difficulty:* Foundation

Worked solution

Step 1. Set up brackets.

$$(2x \quad)(x \quad) = 0$$

$$2x \times x = 2x^2.$$

Step 2. Find a pair multiplying to 5 giving middle term $11x$.

$$(2x + 1)(x + 5) : 10x + x = 11x$$

1 goes with $2x$ and 5 with x .

Step 3. Solve each factor.

$$x = -\frac{1}{2} \text{ or } x = -5$$

Set each factor to zero.

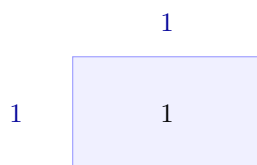
Final answer: $x = -\frac{1}{2}$ or $x = -5$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$1$$

Visualising the factorisation



Area / box model: each cell is the product of its row and column term.

Question 60 qc1-060

Solve by factorising: $3x^2 + 10x - 8 = 0$

Type: Non-monic ($a \neq 1$)

Marks: 3 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(3x \quad)(x \quad) = 0$$

$$3x \times x = 3x^2.$$

Step 2. Opposite signs ($c < 0$). Try pairs of 8 to get $10x$ in the middle.

$$(3x - 2)(x + 4) : 12x - 2x = 10x$$

-2 with $3x$ and $+4$ with x .

Step 3. Solve each factor.

$$x = \frac{2}{3} \text{ or } x = -4$$

Set each factor to zero.

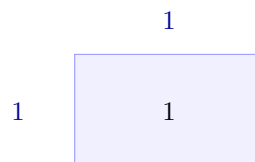
Final answer: $x = \frac{2}{3}$ or $x = -4$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$1$$

Visualising the factorisation



Area / box model: each cell is the product of its row and column term.

Question 61 qc1-061

Factorise: $x^2 + 6xy + 8y^2$

Type: Two variables

Marks: 3 Difficulty: Foundation

Worked solution

Step 1. Set up two brackets, each containing an x term and a y term.

$$(x \quad y)(x \quad y)$$

$x \times x = x^2$ and $y \times y = y^2$ give the end terms.

Step 2. Find numbers multiplying to 8 and adding to 6.

$$2 \times 4 = 8, \quad 2 + 4 = 6$$

Factor pairs of 8: $(1,8), (2,4)$. $(2,4)$ works.

Step 3. Place the numbers beside the y terms.

$$(x + 2y)(x + 4y)$$

Both signs positive.

Final answer: $(x + 2y)(x + 4y)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 4xy + 2xy + 8y^2$$

Visualising the factorisation

	x	$+4y$
x	x^2	$4xy$
$+2y$	$2xy$	$8y^2$

Area / box model: each cell is the product of its row and column term.

Question 62

qc1-062

Factorise: $x^2 + 7xy + 12y^2$

Type: Two variables

Marks: 3 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad y)(x \quad y)$$

Each bracket contains an x and a y term.

Step 2. Find numbers multiplying to 12 and adding to 7.

$$3 \times 4 = 12, \quad 3 + 4 = 7$$

The pair $(3, 4)$ works.

Step 3. Write the answer.

$$(x + 3y)(x + 4y)$$

Both signs positive.

Final answer: $(x + 3y)(x + 4y)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 4xy + 3xy + 12y^2$$

Visualising the factorisation

	x	$+4y$
x	x^2	$4xy$
$+3y$	$3xy$	$12y^2$

Area / box model: each cell is the product of its row and column term.

Question 63

qc1-063

Factorise: $x^2 - 5xy + 6y^2$

Type: Two variables

Marks: 3 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad y)(x \quad y)$$

Standard two-variable setup.

Step 2. Find numbers multiplying to 6 and adding to -5 .

$$(-2) \times (-3) = 6, \quad (-2) + (-3) = -5$$

Both signs negative.

Step 3. Write the answer.

$$(x - 2y)(x - 3y)$$

Both factors negative.

Final answer: $(x - 2y)(x - 3y)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 3xy - 2xy + 6y^2$$

Visualising the factorisation

	x	$-3y$
x	x^2	$-3xy$
$-2y$	$-2xy$	$6y^2$

Area / box model: each cell is the product of its row and column term.

Question 64

qc1-064

Factorise: $x^2 - 4xy - 12y^2$

Type: Two variables

Marks: 3 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad y)(x \quad y)$$

Standard setup.

Step 2. Find numbers multiplying to -12 and adding to -4 .

$$(-6) \times 2 = -12, \quad (-6) + 2 = -4$$

Opposite signs, larger (6) negative.

Step 3. Write the answer.

$$(x - 6y)(x + 2y)$$

Larger factor negative.

Final answer: $(x - 6y)(x + 2y)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 2xy - 6xy - 12y^2$$

Visualising the factorisation

	x	$+2y$
x	x^2	$2xy$
$-6y$	$-6xy$	$-12y^2$

Area / box model: each cell is the product of its row and column term.

Question 65

qc1-065

Factorise: $x^2 - y^2$

Type: Difference of two squares

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Recognise the difference of two squares.

$$x^2 - y^2$$

Both terms are perfect squares separated by a minus sign.

Step 2. Apply $a^2 - b^2 = (a - b)(a + b)$.

$$(x - y)(x + y)$$

Here $a = x$ and $b = y$.

Final answer: $(x - y)(x + y)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + xy - xy - y^2$$

Visualising the factorisation

	x	$+y$
x	x^2	xy
$-y$	$-xy$	$-y^2$

Area / box model: each cell is the product of its row and column term.

Question 66

qc1-066

Factorise: $4x^2 - 25y^2$

Type: Difference of two squares

Marks: 2 *Difficulty:* Foundation

Worked solution

Step 1. Recognise the difference of two squares.

$$4x^2 - 25y^2 = (2x)^2 - (5y)^2$$

Both are perfect squares.

Step 2. Apply the formula.

$$(2x - 5y)(2x + 5y)$$

Here $a = 2x$ and $b = 5y$.

Final answer: $(2x - 5y)(2x + 5y)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$4x^2 + 10xy - 10xy - 25y^2$$

Visualising the factorisation

	$2x$	$+5y$
$2x$	$4x^2$	$10xy$
$-5y$	$-10xy$	$-25y^2$

Area / box model: each cell is the product of its row and column term.

Question 67 qc1-067

Factorise: $x^2 + 13x + 40$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 40 and adding to 13.

$$5 \times 8 = 40, \quad 5 + 8 = 13$$

Factor pairs: $(1,40), (2,20), (4,10), (5,8)$. $(5,8)$ works.

Step 3. Write the answer.

$$(x + 5)(x + 8)$$

Both signs positive.

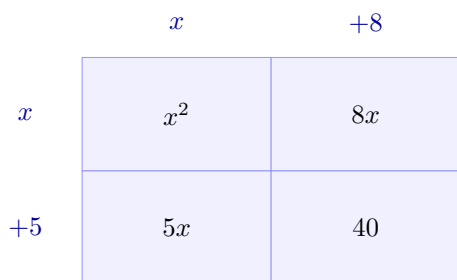
Final answer: $(x + 5)(x + 8)$

Check by expanding

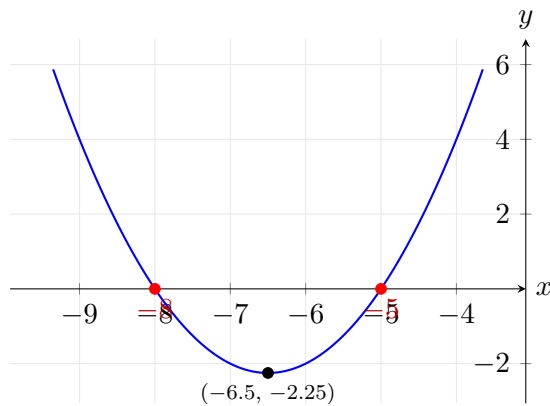
Multiplying the brackets back out reproduces the original expression:

$$x^2 + 8x + 5x + 40$$

Visualising the factorisation



(a) Area / box model



(b) Graph $y = f(x)$ with its roots

Question 68 qc1-068

Factorise: $x^2 + 14x + 45$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 45 and adding to 14.

$$5 \times 9 = 45, \quad 5 + 9 = 14$$

Factor pairs: (1,45), (3,15), (5,9). (5,9) works.

Step 3. Write the answer.

$$(x + 5)(x + 9)$$

Both signs positive.

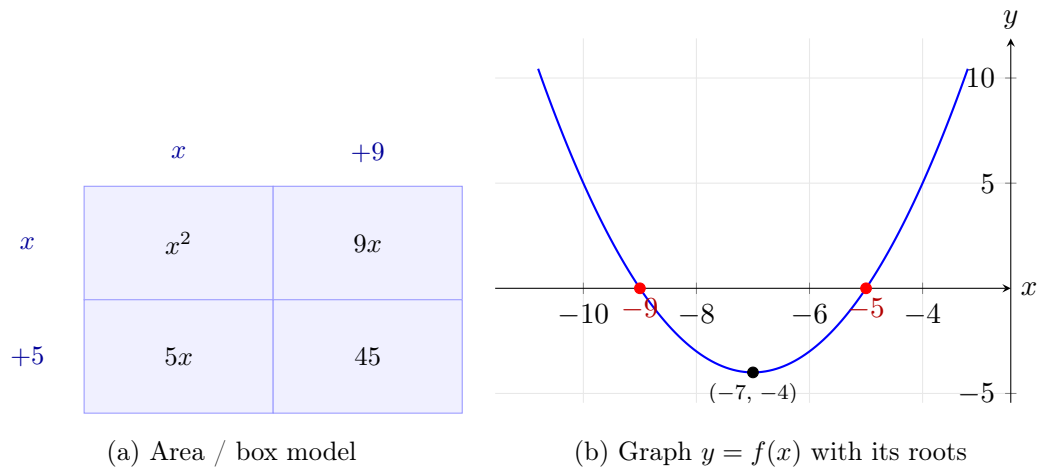
Final answer: $(x + 5)(x + 9)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 + 9x + 5x + 45$$

Visualising the factorisation



Question 69 qc1-069

Factorise: $x^2 - 12x + 32$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to 32 and adding to -12 .

$$(-4) \times (-8) = 32, \quad (-4) + (-8) = -12$$

Both signs negative.

Step 3. Write the answer.

$$(x - 4)(x - 8)$$

Both factors negative.

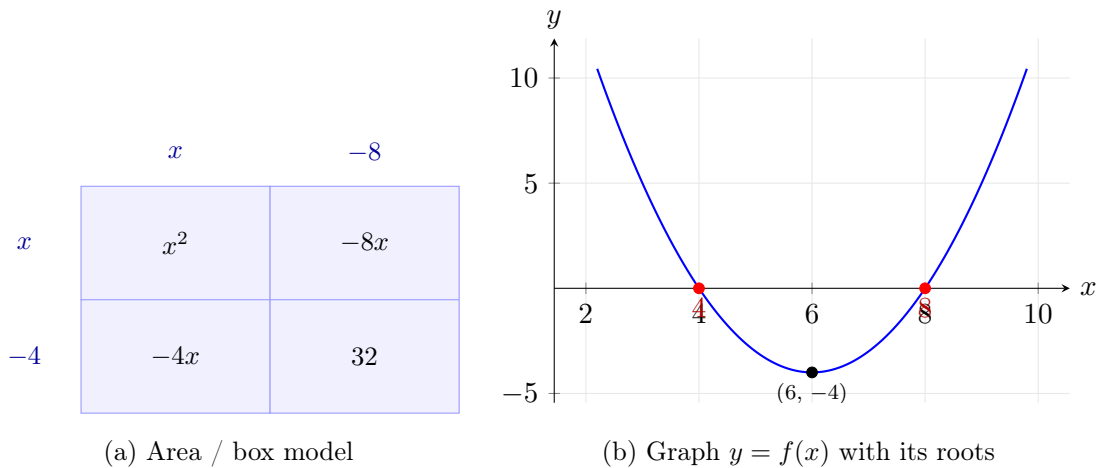
Final answer: $(x - 4)(x - 8)$

Check by expanding

Multiplying the brackets back out reproduces the original expression:

$$x^2 - 8x - 4x + 32$$

Visualising the factorisation



Question 70 qc1-070

Factorise: $x^2 + 6x - 40$

Type: Monic ($a = 1$)

Marks: 2 Difficulty: Foundation

Worked solution

Step 1. Set up brackets.

$$(x \quad)(x \quad)$$

Monic quadratic.

Step 2. Find numbers multiplying to -40 and adding to $+6$.

$$10 \times (-4) = -40, \quad 10 + (-4) = 6$$

Opposite signs; larger factor (10) positive.

Step 3. Write the answer.

$$(x + 10)(x - 4)$$

Larger positive.

Final answer: $(x + 10)(x - 4)$

Check by expanding

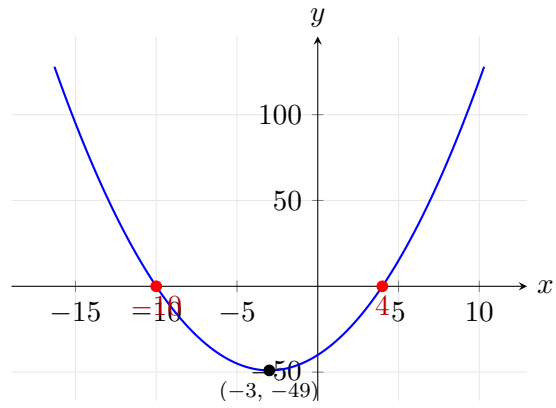
Multiplying the brackets back out reproduces the original expression:

$$x^2 - 4x + 10x - 40$$

Visualising the factorisation

	x	-4
x	x^2	$-4x$
$+10$	$10x$	-40

(a) Area / box model



(b) Graph $y = f(x)$ with its roots